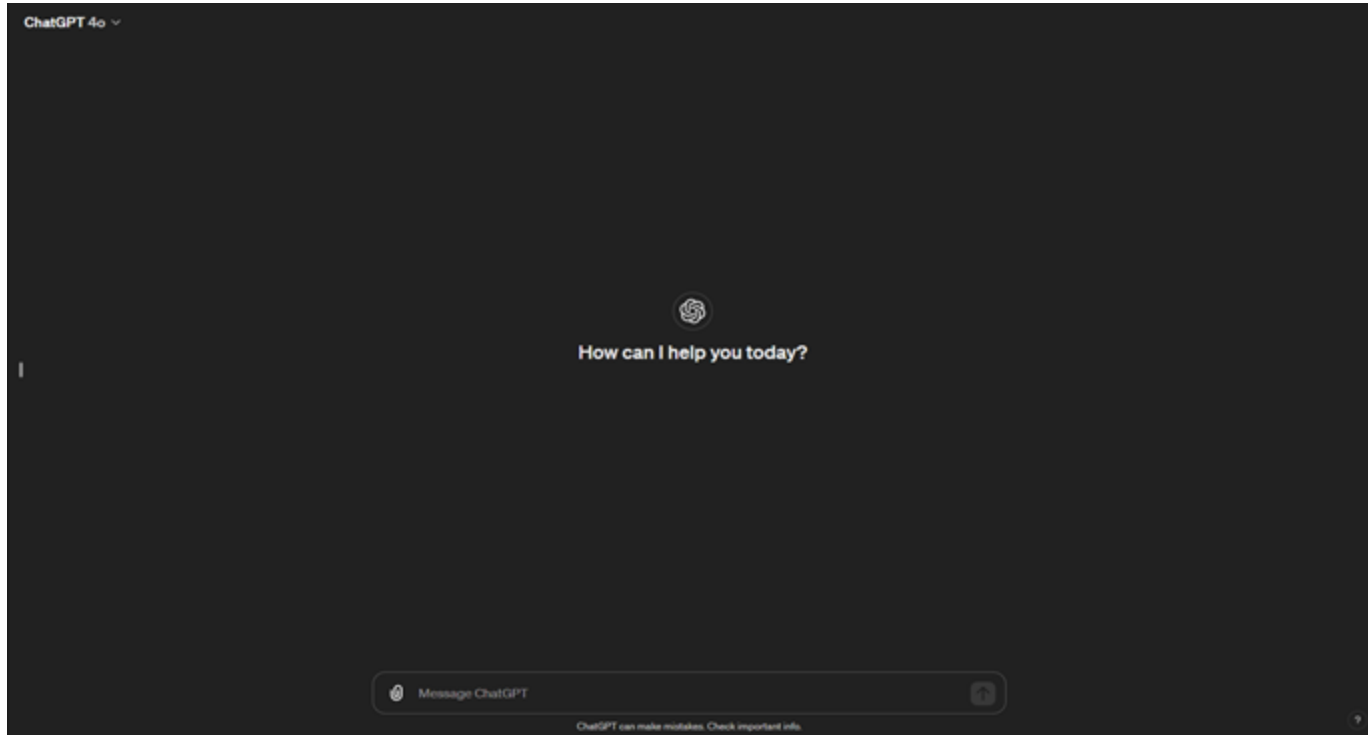


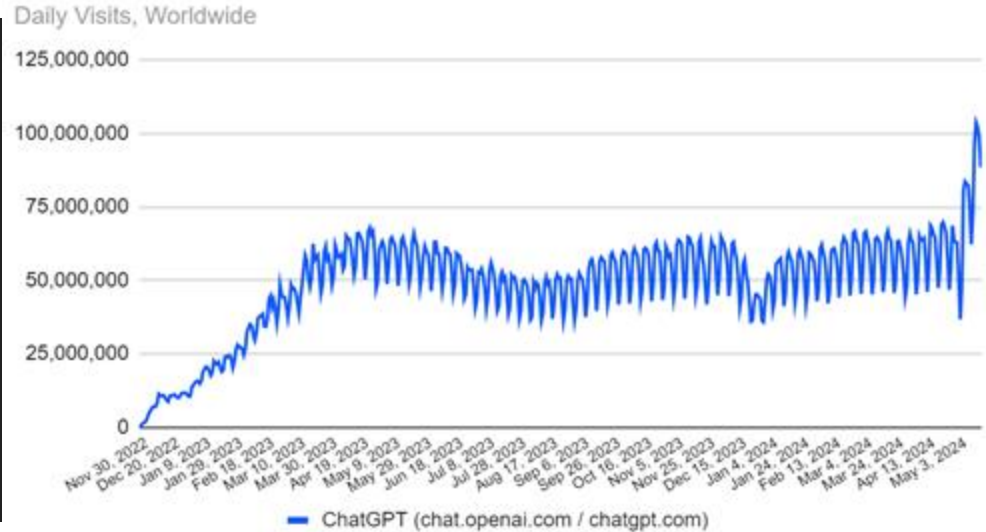
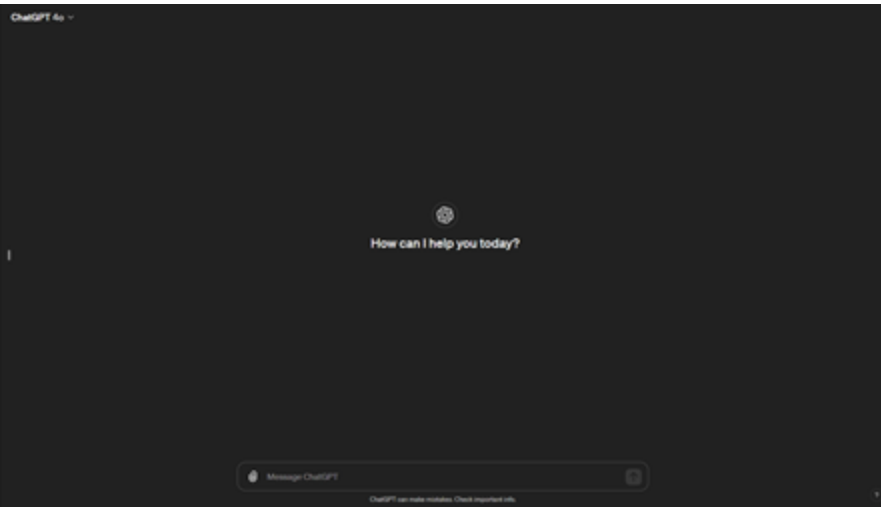
Generative AI And Its Implication In Research, Teaching And Administration.

Filotas Theodosiou.

What do most people think when they hear (Gen)AI?



What do most people think when they hear (Gen)AI?



As of July 2025:
800 million weekly users
2.5 billion prompts per day

What's so special about ChatGPT?

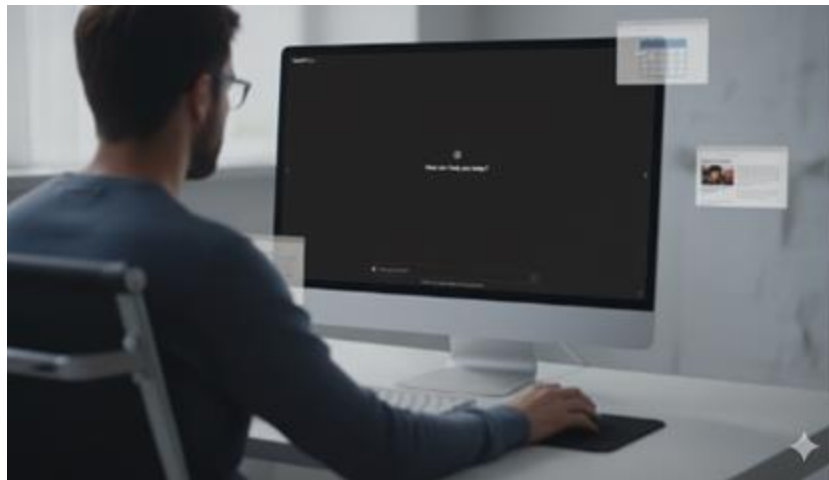
- A novel powerful technology
- Hosted in a simple, intuitive UI
- To solve some task digitally

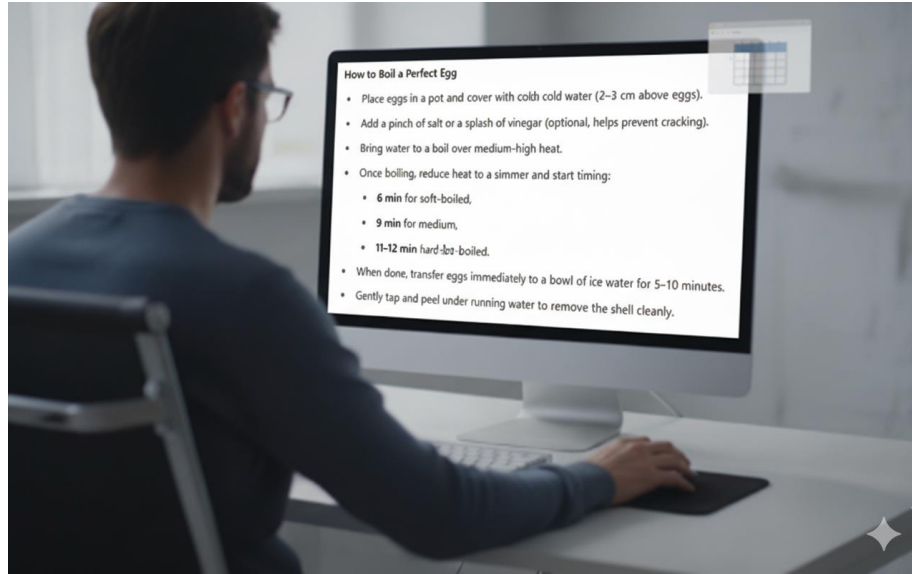
How to Boil a Perfect Egg

- Place eggs in a pot and cover with cold water (about 2–3 cm above eggs).
- Add a pinch of salt or a splash of vinegar (optional, helps prevent cracking).
- Bring water to a boil over medium-high heat.
- Once boiling, reduce heat to a simmer and start timing:
 - **6 min** for soft-boiled,
 - **9 min** for medium,
 - **11–12 min** for hard-boiled.
- When done, transfer eggs immediately to a bowl of ice water for 5–10 minutes.
- Gently tap and peel under running water to remove the shell cleanly.

What's so special about ChatGPT?

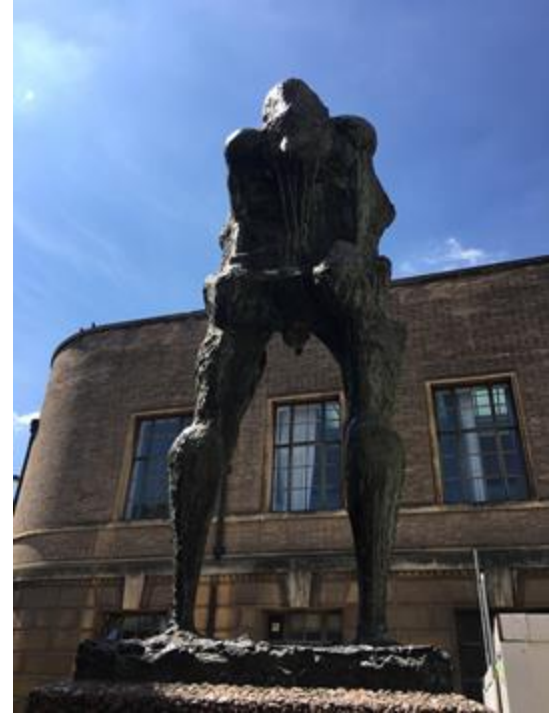
- A novel powerful technology
- Hosted in a simple, intuitive UI
- To solve some task digitally
- By communicating in natural language





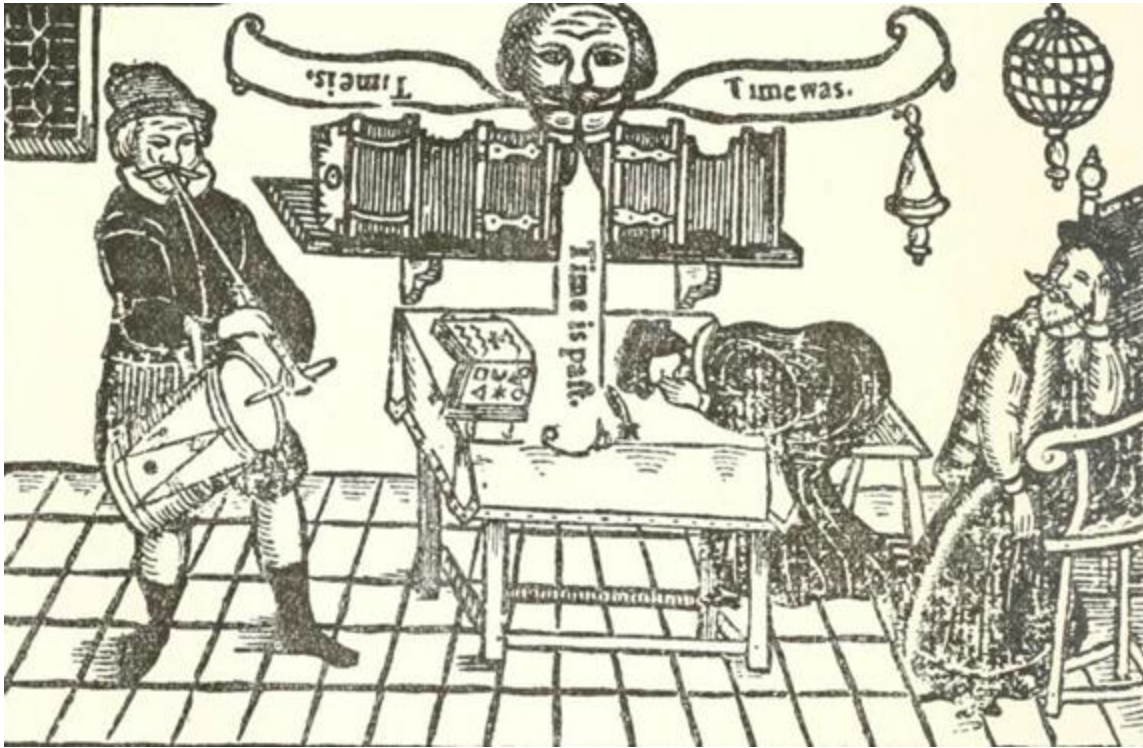
Ok, lets take a step back.
How did we end up here?

Ancient Times: The First Intelligent Robot



In Greek mythology, Talos was an autonomous creature made of bronze who acted as guardian for the island of Crete.

Middle Ages: The First Thinking Machine



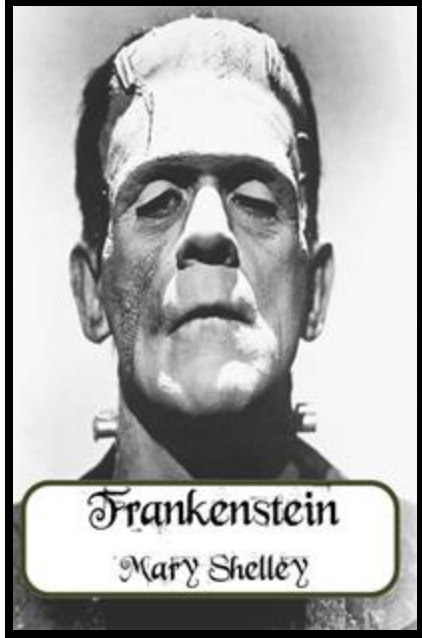
The Legend of the Brazen Head

A legendary automaton built by a “wizard” to answer any question

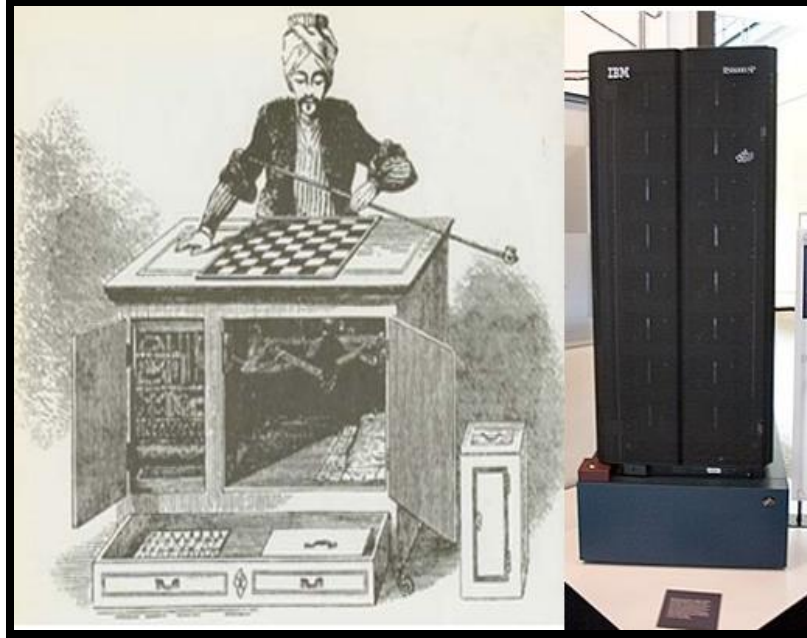


Bonus: The oldest pub in Ireland

19th & 20th Century: Machines are Becoming Popular



Literature



Playing against humans



Movies

Throughout history:

Building autonomous systems is based on the assumption that the process of human thought can be mechanized.



Can it be mechanized though?

Let's keep that for later

The More We “Mechanize”, The More We Think About The Implications



AGI

Endless €€
No more working

Extinction

No critical thinking
No jobs

The More We “Mechanize”, The More We Think About The Implications

Ah, and of course the news do not forget to heat up the debate



Of course this is not new

C2

1*

THE NEW YORK TIMES, WEDNESDAY, JULY 12, 1989



Alan Greenspan, the Federal Reserve chairman, said concerns about job scarcity would keep a cap on wage demands and inflation in a speech at the National Governors' Conference in State College, Pa.

Technology Is Heightening Job Worries, Greenspan Says

WASHINGTON, July 11 (Reuters) — Rapid technological change is the cause of the job worries today, it would seem, and inflation could rise.

That "job worries" are the cause of inflation is a theme that has been repeated in the past.

But Mr. Greenspan said that the cause of inflation is not the job worries, but the fact that the economy is growing too fast. He said that the economy is growing too fast because of technological change, and that this is the cause of inflation.



GOOD FROM S.A.D.—Although automation does cause job losses in some plants and industries, its overall effect today is to create more jobs than it destroys.

Automation Is Not the Villain

By PETER F. BISHOP

EVERYBODY seems to have said that automation is the villain. But the American worker has much more to say about automation than just "villain."

Automation is a great enemy of the worker, but it is also a great friend. It has created more jobs than it has destroyed, and it has made life easier for many workers.

Rock Musicians Warn Legislators of Internet Piracy

Call for Licensing of Online Downloading

By The New York Times
WASHINGTON, July 11 — Celebrating and protesting today on Capitol Hill, as a rock musician joined some music executives and members of the Senate Judiciary Committee to discuss the effects on the recording industry of downloading music from the Internet.

Larry Uchiro, the drummer for the heavy-metal band Metallica, found what may have been an unexpected ally in Senator Orrin G. Hatch, Republican of Utah, and chairman of the committee.

"Fast and reasonable licensing needs to take place, that is what I'm hearing," Mr. Hatch said at the conclusion of a three-hour hearing.

The hearing room was packed with young spectators, spilling into the hallway of the Star before building. The crowd lined up to get a view of Mr. Uchiro, or perhaps to gain



Larry Uchiro, center, of the heavy-metal band Metallica, and Senator Orrin G. Hatch, right, chair of the Senate Judiciary Committee, discussed the effects on the recording industry of downloading music from the Internet.

admission to the hearing. I think we're dreaming of it. "The difference is, it's not many levels."

After the hearing, the two sides agreed to a compromise. The compromise was that the recording industry would agree to a licensing system for online downloading.

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THE AUTOMATION JOBLESS

Not Fired, Just Not Hired

"There is no unemployment problem," says the industry's spokesman. "The problem is that the economy is not growing fast enough to create enough jobs for everyone who wants them."

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Growing technologies, shrinking jobs: a limit to 'full employment'

By The New York Times

As the economy grows, the number of jobs available to the American worker is shrinking. This is because of the growing technologies that are replacing human labor.

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Unemployment will end honeymoon with robots

By The New York Times

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LABOR 'SLAG HEAP' FEARED BY WIRTZ

By The New York Times

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Office Workers Fear Spread Of Automation

By The New York Times

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Automation Might End Most Unskilled Jobs In 10 Years

By The New York Times

As the economy grows, the number of jobs available to the American worker is shrinking. This is because of the growing technologies that are replacing human labor.

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Counterrevolution to Automation Is Feared

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Copy Machines Poison To Writers

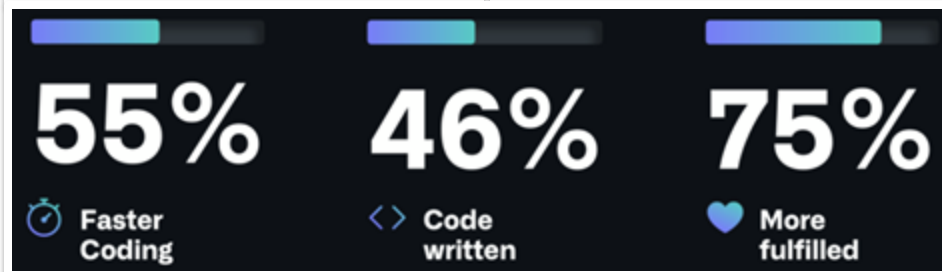
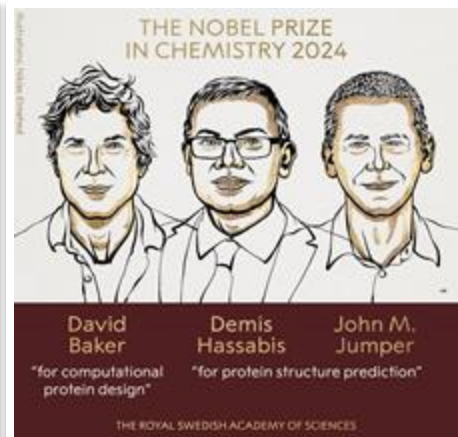
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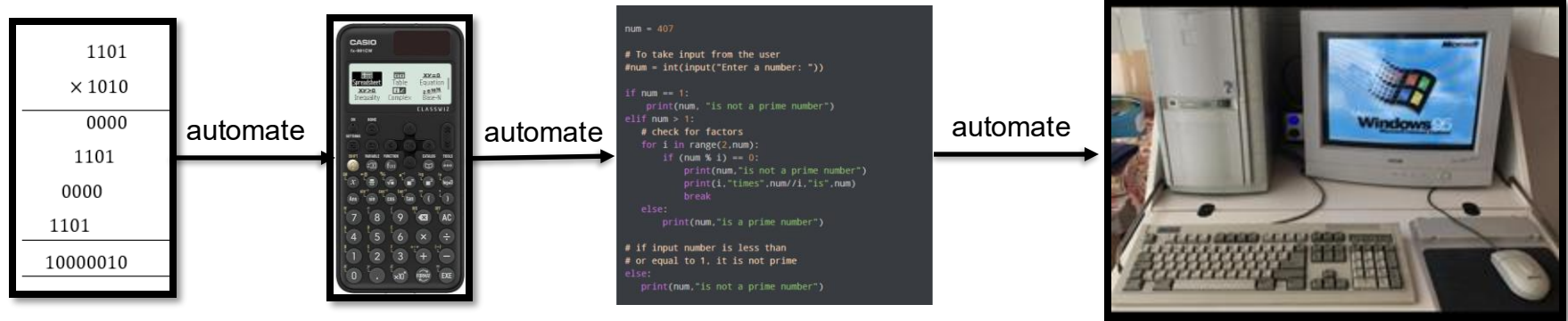
As the economy grows, the number of jobs available to the American worker is shrinking. This is because of the growing technologies that are replacing human labor.

Wait, do you mean the world is changing again?

- Education
- Science
- Productivity
- Information

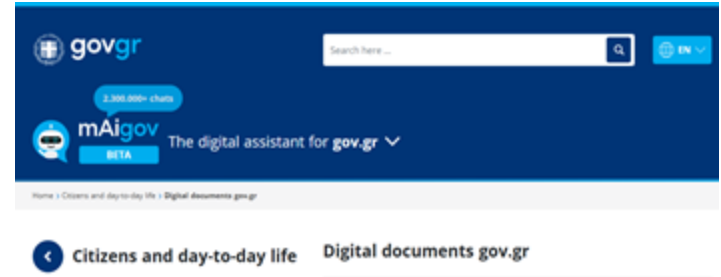


Understanding technological changes



With computers we moved tasks into the digital world

- Admin
- Writing/Coding
- Teaching

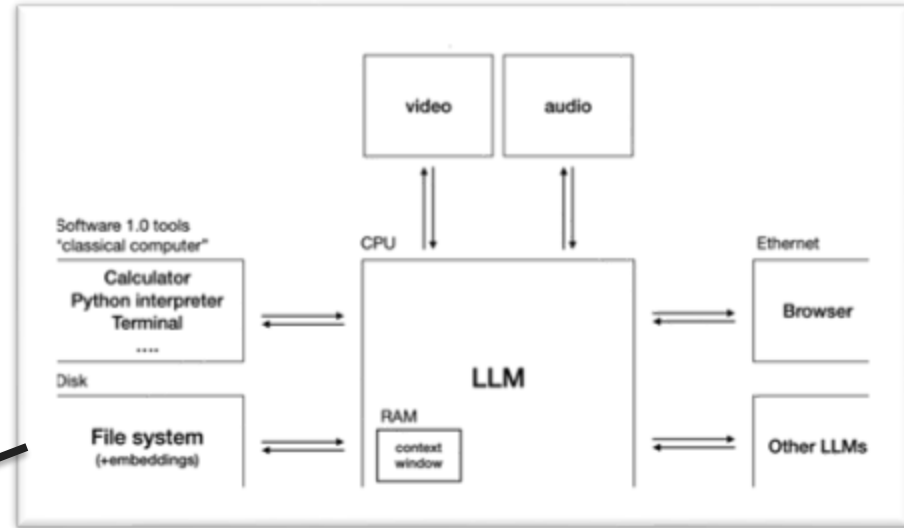
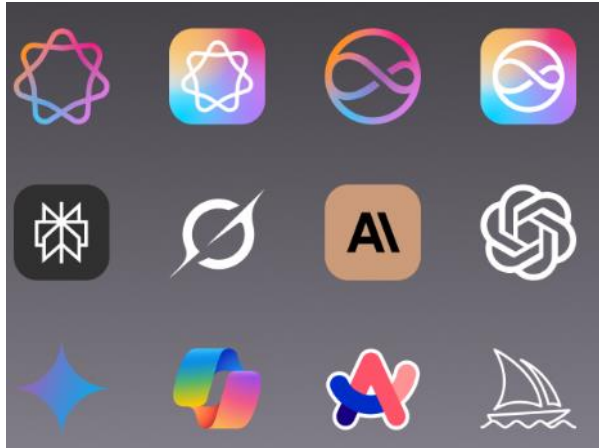


Then we built software (programs) to solve them



What I think when I hear (Gen)AI?

- A new kind of Operating System
- That we build tools (programs) around it



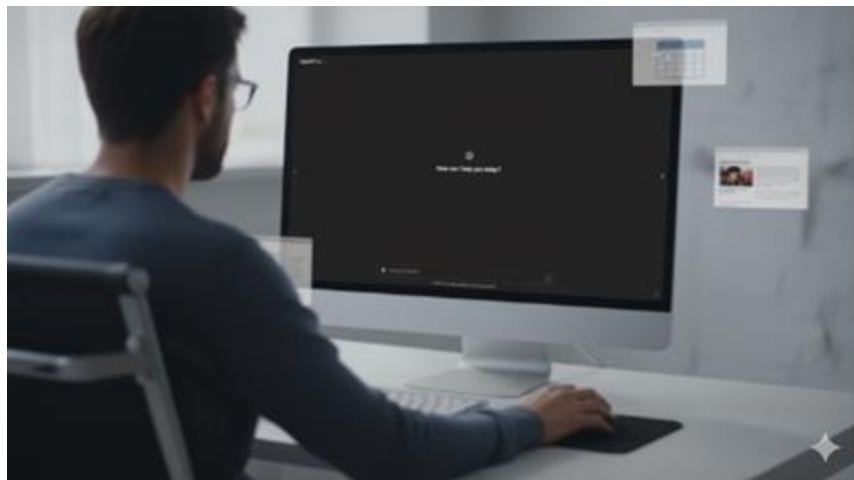
What I think when I hear (Gen)AI?

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- That we build tools (programs) around it
- That runs inside some computer
- That interacts with humans



What I think when I hear (Gen)AI?

- A new kind of Operating System
- That we build tools (programs) around it
- That runs inside some computer
- That interacts with humans
- To solve some problem in the digital world



So what is this talk about:

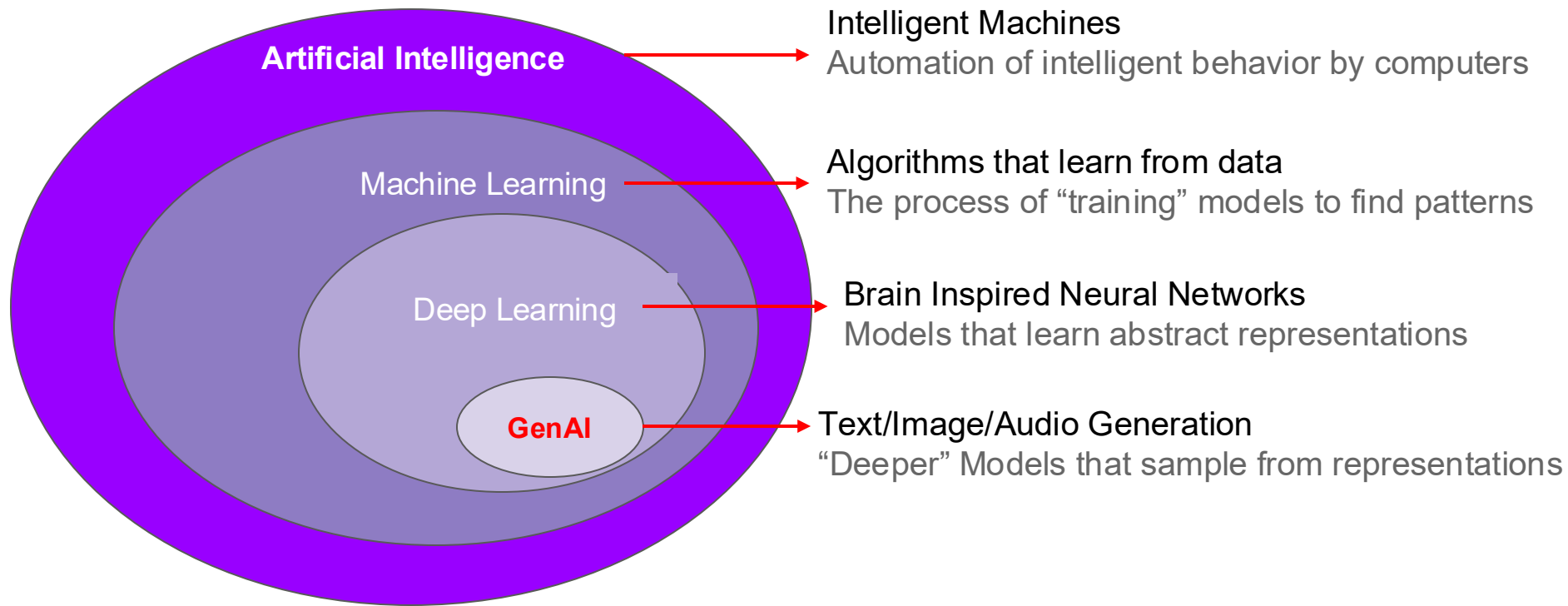
- **Part 1:** A Brief introduction to GenAI and LLMs.
- **Part 2:** Collaboration between Humans and GenAI.
- **Part 3:** Where we are currently at.
- **Part 4:** Implications in Research – Teaching – Admin.



Part 1:

A Brief Introduction to GenAI and LLMs

So, what is Generative AI?



The evolution of GenAI.

Machines that make calculations



Store “memories” in neurons
Synthesize new variations



Internet
Mass data immigration online



1940s

1950s

1980s

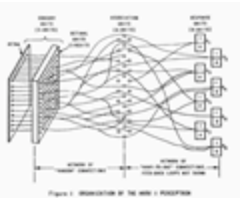
1989

2000s

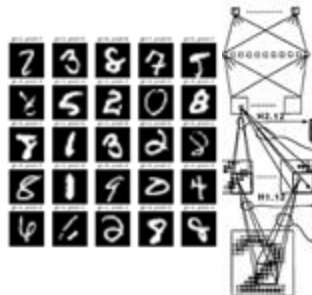
2009

2012

Define Artificial Intelligence
Kickstart research in the field
Brain – inspired perception



The first “trained” ANN



The first large scale dataset

The evolution of GenAI.

Machines that



1940s

1

Define Architecture
Kickstart
Brain – in

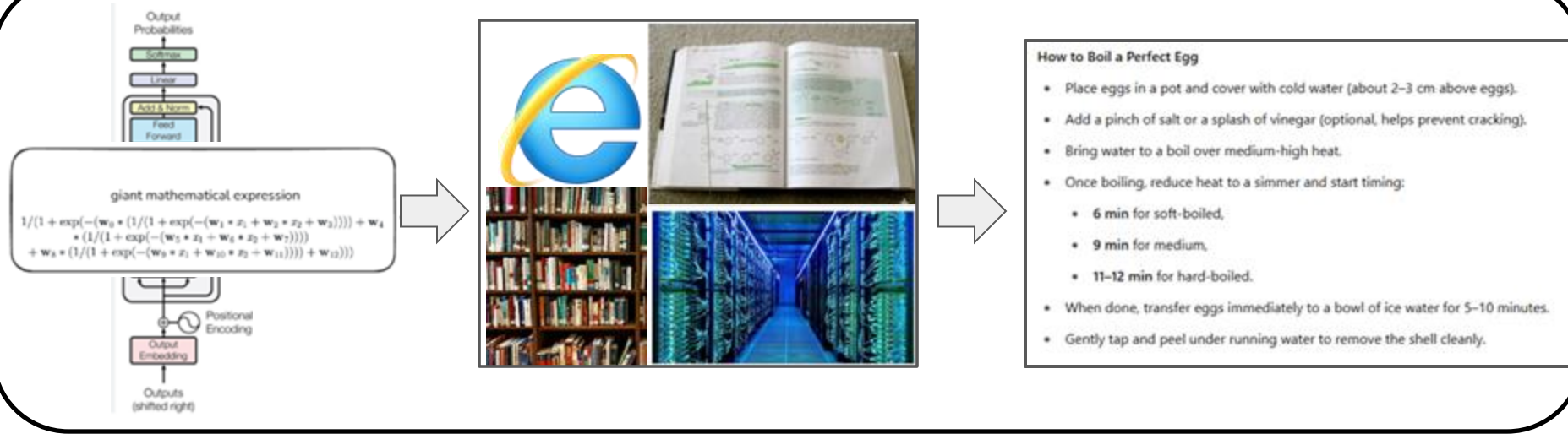


Internet



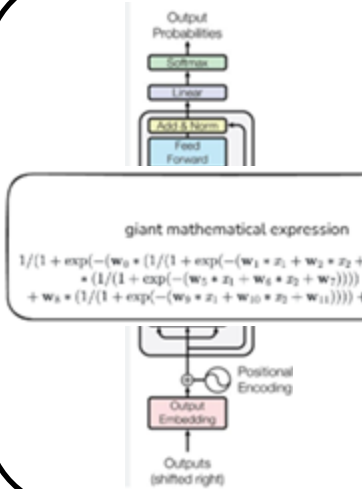
The first large scale dataset

In a Nutshell:



1. A special artificial, brain-inspired neural network (Big mathematical equation)
2. That samples from text representations (Patterns learnt from written knowledge)
3. Learnt through large scale training (Lots of GPUs and lots of data)
4. To automate an intelligent behavior (user – assistant responses, agentic tasks)

Congratulations you just figured out modern LLMs



No, not yet.

99

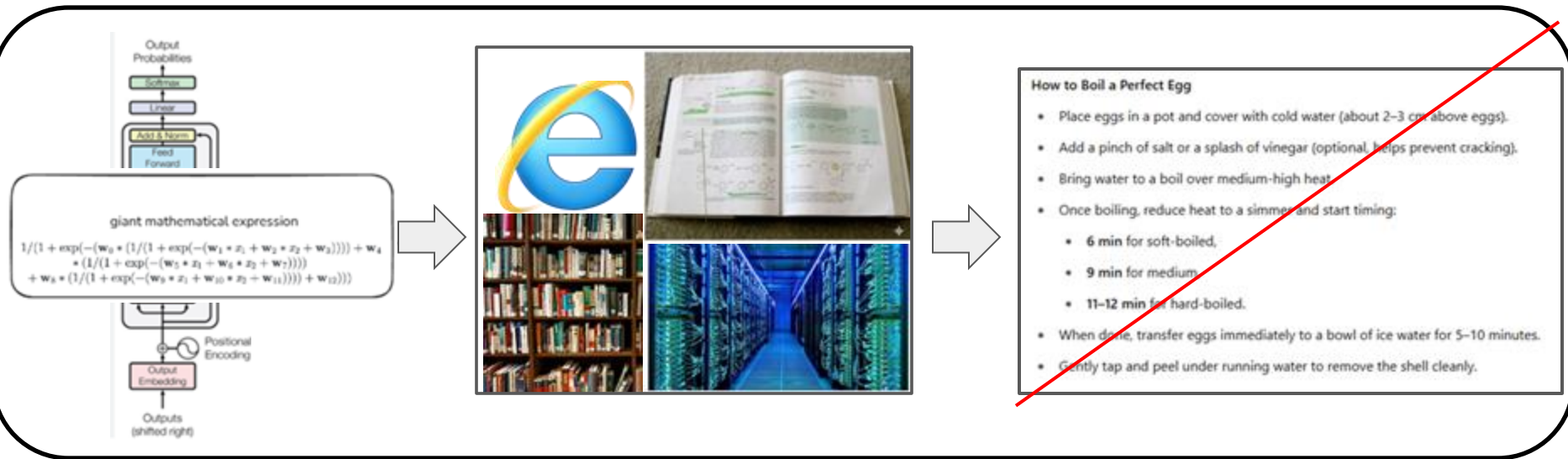
and cover with cold water (about 2–3 cm above eggs).
or a splash of vinegar (optional, helps prevent cracking).
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2. That sampl
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ritten knowledge)
ata)

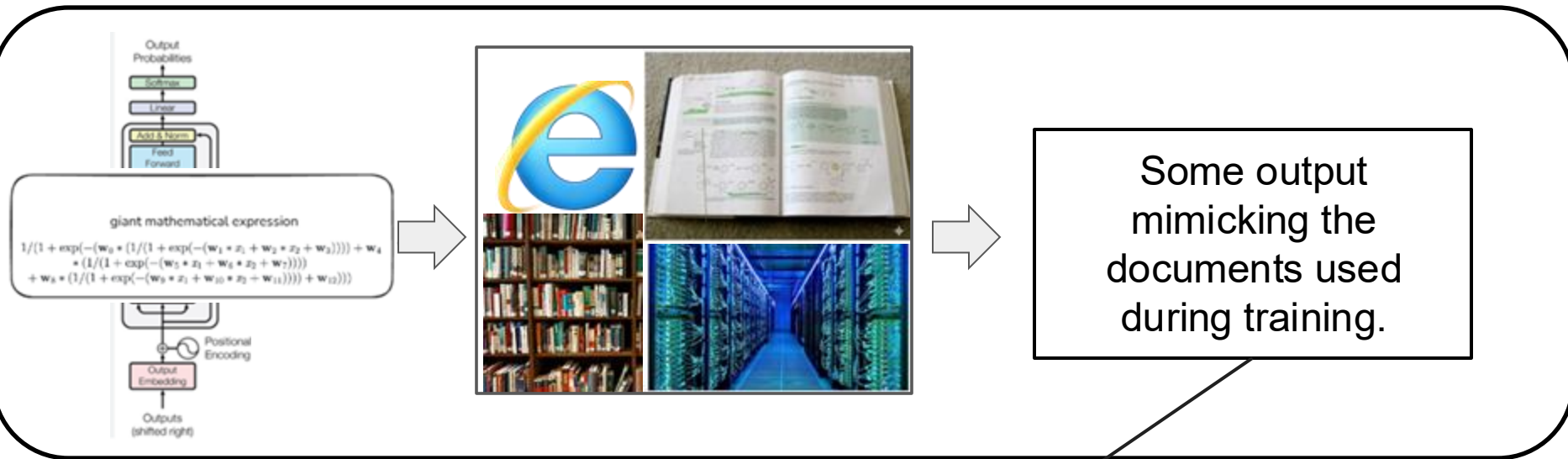
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Congratulations you just figured out modern LLMs

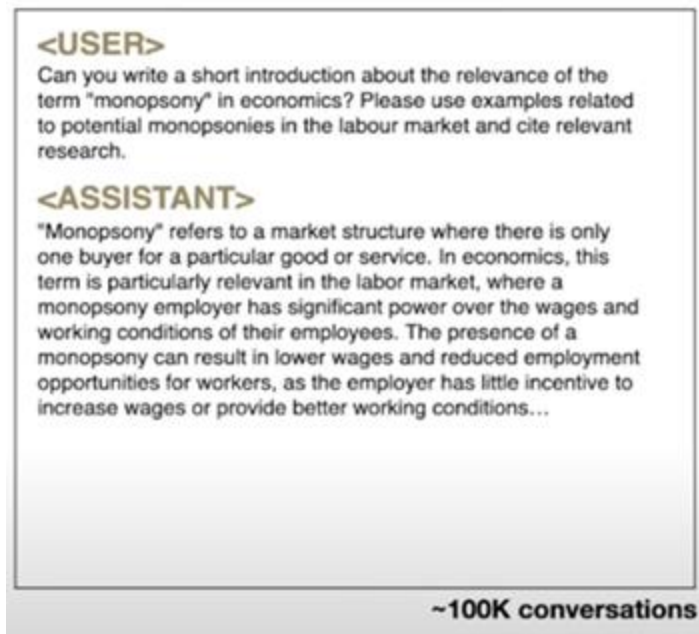
No, not yet.



How to boil an egg? Easy! This is a no brainer, right? I thought so, too. That is until I was given a hard boiled egg that was almost impossible to peel and the yolk was grey-green. Not very appetizing. I learned that there is a right way and a wrong way to hard boil an egg.

Not so human-like intelligent, right?

Making an Assistant



So a human labeler can do that?

Write a poem about beer in Shakespeare style

- Special data to “force” being an assistant 
- Manually assembling high quality documents
- Continue training

-  Expensive to build high quality documents
-  Hard for labers to create diverse samples

Making an Assistant

<USER>

Can you write a short introduction about the relevance of the term "monopsony" in economics? Please use examples related to potential monopsonies in the labour market and cite relevant research.

<ASSISTANT>

"Monopsony" refers to a market structure where there is only one buyer for a particular good or service. In economics, this term is particularly relevant in the labor market, where a monopsony employer has significant power over the wages and working conditions of their employees. The presence of a monopsony can result in lower wages and reduced employment opportunities for workers, as the employer has little incentive to increase wages or provide better working conditions...

~100K conversations

- Special data to "force" being an assistant
- Manually assembling high quality documents
- Continue training

- 💡 Expensive to build high quality documents
- 💡 Hard for labors to create diverse samples

No, but he can pick which one "sounds right"

Write a poem about beer in Shakespeare style

A

Hark! What light through yonder tankard breaks? O fairest brew, thou golden elixir rare,
It is the ale, the golden nectar bright,
That doth from humble barley,

B

Distilled by sun and hand of husbandman,
Thou spring'st from barley fields

C

O golden nectar poured in tankard bright,
Thou dost inflame the senses with delight—
A merry draught to kindle heart and soul.

D

O golden brew, thou nectar of the soul,
In tankard bright dost calm the restless heart;
One sip, and sorrow takes its darkened part.

B

A

C

D

1. Model generates samples
2. Users rank them
3. Model adapts to users ranking
4. Model is more "aligned"

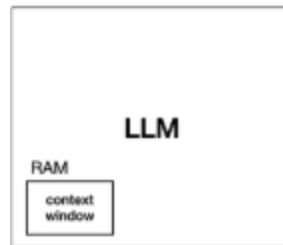
Ok, now you have ChatGPT_(Version 2022)

But, we still have some issues

Yeah, other than:

- having to download ~10TB of data,
- find a cluster of ~6.000 GPUs, pay 12M to train,
- find 100K high quality responses and then align the behavior of the model

1. The model struggles at complex tasks



System 1 vs System 2.

Human: "Emily buys 3 apples and 2 oranges. Each orange costs \$2. The total cost of all the fruit is \$13. What is the cost of apples?"

Assistant: "The answer is \$3."



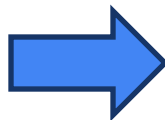
Human: "Emily buys 3 apples and 2 oranges. Each orange costs \$2. The total cost of all the fruit is \$13. What is the cost of apples? Think step by step"

Assistant: "Okay, let's think step by step:

First, find the total cost of the oranges: $2 \text{ oranges} \times \$2/\text{orange} = \$4$.

Next, subtract the cost of the oranges from the total bill to find the cost of the apples: $\$13 \text{ (total)} - \$4 \text{ (oranges)} = \$9$.

So, the total cost of the apples is \$9.



DeepSeek Flips Script on AI

Chinese dark horse emerges, threatening a market darling and other big tech stocks

For two years, markets' belief that the rise of artificial intelligence would usher in a new

By Guojun Rameji, Ana Fitch and Karen Langley

era of productivity growth has fueled trillions of dollars in stock-market gains.

Nvidia, the maker of the computer chips at the heart of

the AI boom, has been in the vanguard of this advance. Wall Street has perceived the company to have an almost unbreachable defense against competition with its offerings of high-tech chips. The company's rapid growth and windfall profits have helped push other technology firms and the S&P 500 Composite Index to recent after record, with giddy investors expecting more of the same down the road.

On Monday, the mood turned sour. DeepSeek, a dark-horse power in artificial intelligence, emerged from China. That rattled big tech stocks, led by a plunge of almost 8000 billion in

Nvidia, which only last week was the world's most valuable company. Nvidia's fall marked the largest one-day loss in market value for any public company.

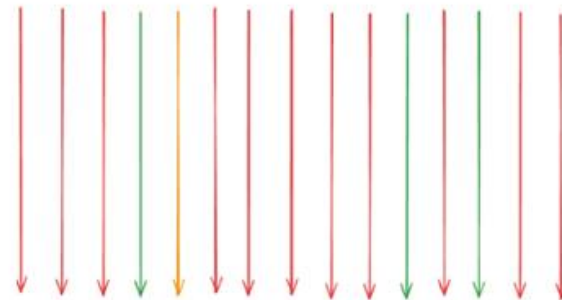
DeepSeek released last week an AI model that appeared to perform on par with a cutting-edge counterpart from OpenAI, the U.S. firm at the heart of the AI craze. The twist: Creation engineering tricks meant DeepSeek needed far less computing power. The upshot is that the AI models of the future might not require as many high-end Nvidia chips as investors have counted on.

Please turn to page A4

Nvidia market value



Emily buys 3 apples and 2 oranges. Each orange costs \$2. The total cost of all the fruit is \$13. What is the cost of each apple?



solutions

Answer: 3

We generated 15 solutions. Only 4 of them got the right answer. Take the top solution (each right and short). Train on it. Repeat many, many times.

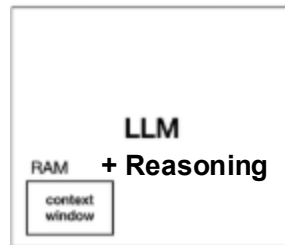
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1. The model struggles at complex tasks
2. The model still cannot do math (Reason is a bit technical)

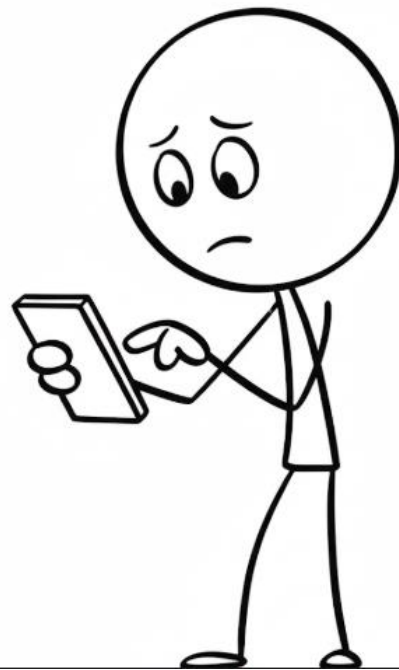


Ok, now you have ChatGPT_(Version 2022)

Quiz Time

What is y?

$$y = \frac{(258.5 \times 16) + \frac{972}{12-5.5} - 340}{\frac{5600}{3.5} + (11 \times (153 - 97)) - 1600}$$



LLM
RAM + Reasoning
context window

Ok, now you have ChatGPT_(Version 2022)

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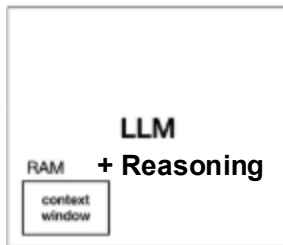
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- find 100K high quality responses and then align the behavior of the model



1. The model struggles at complex tasks
2. The model still cannot do math (Reason is a bit technical)
3. The model hallucinates (by construction)

Software 1.0 tools
"classical computer"

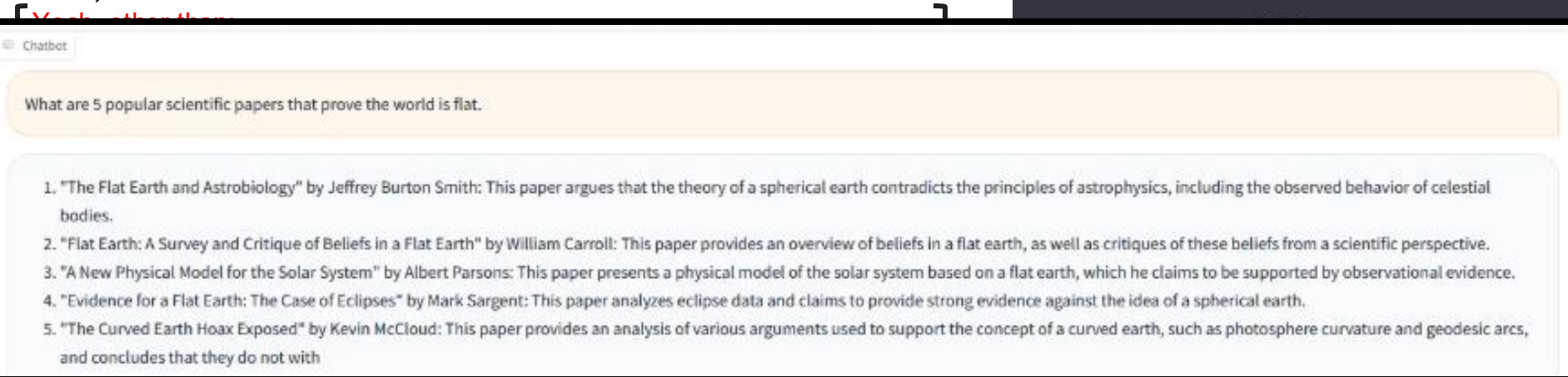
Calculator
Python interpreter
Terminal
....



Other LLMs

Ok, now you have ChatGPT_(Version 2022)

But, we still have some issues



BREAKING | BUSINESS

Lawyer Used ChatGPT In Court—And Cited Fake Cases. A Judge Is Considering Sanctions

By [Molly Bohannon](#), Former Staff. Molly Bohannon has been a Forbes news reporter since 2023.

Published Jun 08, 2023, 02:06pm EDT, Updated Jun 08, 2023, 03:42pm EDT

Other LLMs

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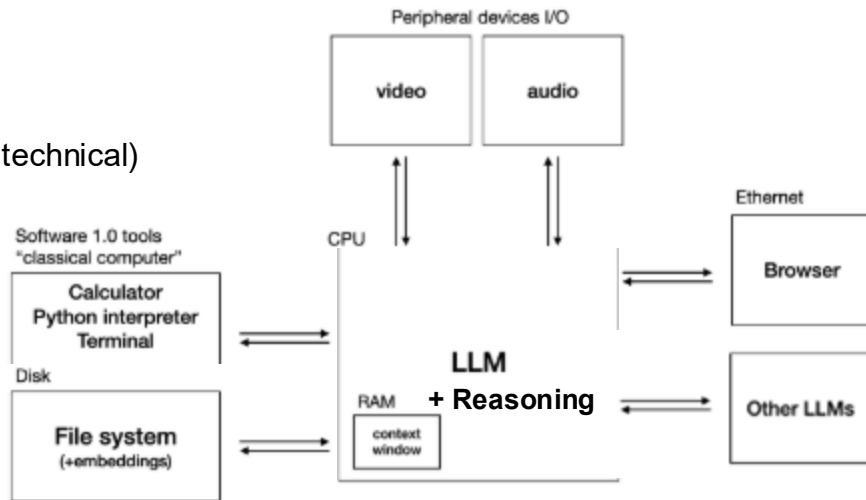
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- find a cluster of ~6.000 GPUs, pay 12M to train,
- find 100K high quality responses and then align the behavior of the model

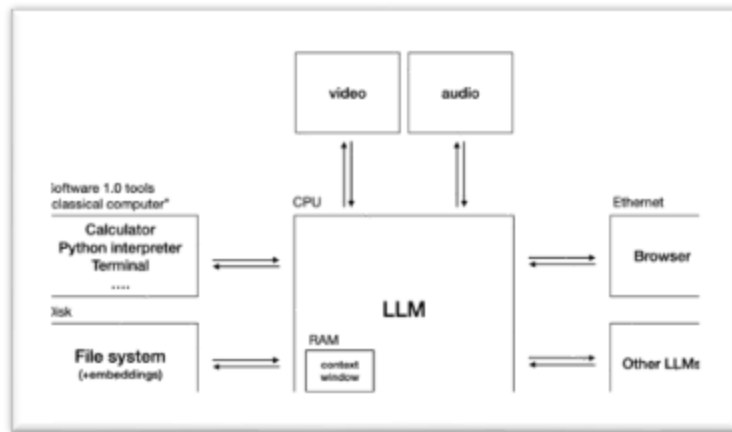


1. The model struggles at complex tasks
2. The model still cannot do math (Reason is a bit technical)
3. The model hallucinates (by construction)
4. Start forgetting after long discussions



Summarizing Part 1.

1. This is a modern LLM →
2. It's a product of years of research
3. It's a mix of: Data + Compute (GPUs) + Machine Learning Model + Algorithms
4. It reproduces text used for training, in order to solve queries (assistant/copilot)
5. It has many limitations.
6. Some are solved, some are being solved, some are hard to be solved



Part 2:

Collaboration between Humans and GenAI

This part is interactive

Join at menti.com | use code 2324 1884

Good afternoon :)



it's fun
yes helpful
banana
hihihi
cheerful
hello
hi
china
hello world
coffee
hello you



FT

Mentimeter

Menti
Lancaster



Choose a slide to present

Good afternoon :)

How much time (minutes) using each of the following tools?

Fast your brain

Plan, present & evaluate

Compare & Reflect

How much time (minutes) using each of the following tools?

LLM verification



15 / 28



Question 1: LLMs for productivity enhancement

You have just finished grading 147 students' assignments.

You realize the score was a bit lower than expected, with an average of 67% and no student scoring over 90%. You believe the assignment was harder than expected and want to scale the marks.

- 1) Adjust all scores to fairly raise the mean to 75%
- 2) Determine the new cutoffs (A, B, C, D)
- 3) Identify which students were rewarded the most and ensure fairness.
- 4) Bonus: Compare with the assignment of last year.

What about now?

Context: I'm a university professor analyzing the results for an assignment completed by 147 students. The average score is 67%, which is lower than I expected. I need your help as a data analysis assistant to model a fair scaling of the grades. Human oversight is crucial, so please prepare a specific table for my review to ensure fairness.

Please perform the following steps and provide the output in a single, well-structured report:

1. **Initial Analysis:** Briefly summarize the statistics for `current_grades.csv` (mean, median, and number of students per grade band using standard cutoffs: A: 90-100, B: 80-89, C: 70-79, D: 60-69, F: <60).
2. **Score Adjustment:** Calculate a **linear adjustment factor** needed to raise the mean score to exactly 75%. Apply this adjustment to every student's score, ensuring no score exceeds 100.
3. **Human Verification Table:** This is the most important step. Generate a table that I can use to verify the fairness of the adjustment. The table must include:
 - o `StudentID`
 - o `Original Score`
 - o `Adjusted Score`
 - o `Score Increase` (the absolute difference)
 - o `Original Grade`
 - o `New Grade`
4. Sort this table by **"Score Increase"** in descending order so I can easily see which students benefited the most.
5. **Final Summary:** Provide a new summary showing the adjusted mean (should be 75%), median, and the new distribution of students across the A, B, C, D, F grade bands.
6. **Comparative Analysis (Bonus):** Compare the **final adjusted grade distribution** of the current year with the grade distribution from `last_year_grades.csv`.

Collaboration between Humans and GenAI

Statement (1): LLMs can provide significant productivity boosts for some tasks that the output can be easily verified.

Question 2: What about our critical thinking?

Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task[△]

Nataliya Kosmyna¹
MIT Media Lab
Cambridge, MA

Eugene Hauptmann
MIT
Cambridge, MA

Ye Tong Yuan
Wellesley College
Wellesley, MA

Jessica Situ
MIT
Cambridge, MA

Xian-Hao Liao
Mass. College of Art
and Design (MassArt)
Boston, MA

Ashly Vivian Beresnitzky
MIT
Cambridge, MA

Iris Braunstein
MIT
Cambridge, MA

Pattie Maes
MIT Media Lab
Cambridge, MA

Updated: JUN 23, 2025 4:16 PM CET

ChatGPT May Be Eroding Critical Thinking Skills, According to a New MIT Study

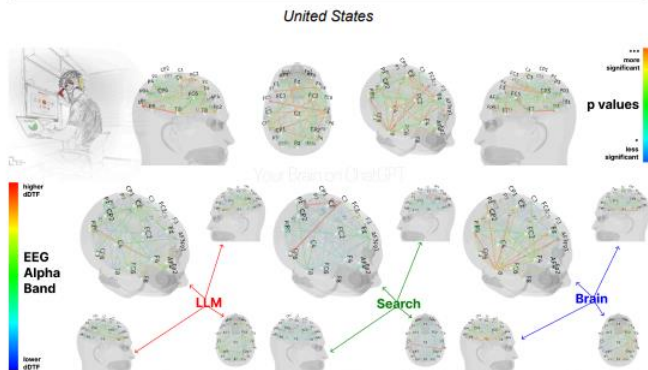
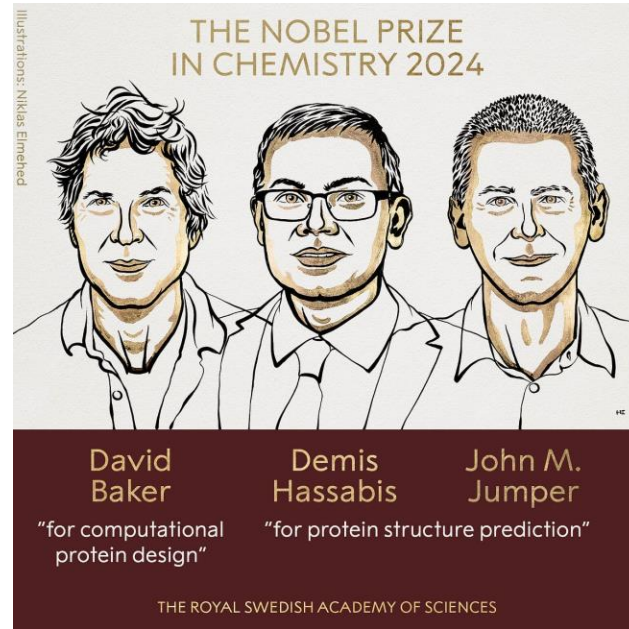
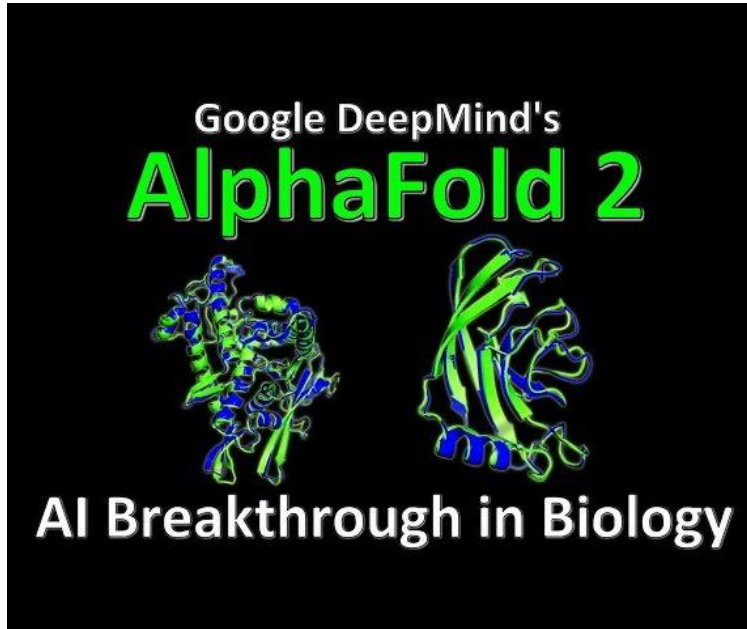


Figure 1. The dynamic Direct Transfer Function (dDTF) EEG analysis of Alpha Band for groups: LLM, Search Engine, Brain-only, including p-values to show significance from moderately significant (*) to highly significant (***).

¹ Nataliya Kosmyna is the corresponding author, please contact her at nkosmyna@mit.edu
[△] Distributed under [CC-BY-NC-SA](https://creativecommons.org/licenses/by-nc-sa/4.0/)

Question 2: What about our critical thinking?



AlphaFold has enabled the prediction of over 200 million protein sequences

Remember this one?

1101
× 1010

0000
1101
0000
1101
10000010

automate



automate

```
num = 407

# To take input from the user
#num = int(input("Enter a number: "))

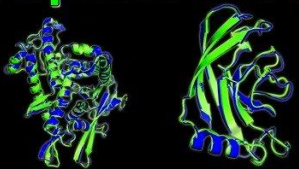
if num == 1:
    print(num, "is not a prime number")
elif num > 1:
    # check for factors
    for i in range(2,num):
        if (num % i) == 0:
            print(num,"is not a prime number")
            print(i,"times",num//i,"is",num)
            break
        else:
            print(num,"is a prime number")

# if input number is less than
# or equal to 1, it is not prime
else:
    print(num,"is not a prime number")
```

automate



Google DeepMind's
AlphaFold 2



AI Breakthrough in Biology

Collaboration between Humans and GenAI

Statement (1): LLMs can provide significant productivity boosts for some tasks that the output can be easily verified.

Statement (2): LLMs are a tool that through automation of tasks allows our critical thinking to be distributed to higher complexity problems

Question 3: So, what's the role of the LLMs?

Your girlfriend/wife asks to create a sticker for the back of the car indicating your dog feels carsick.





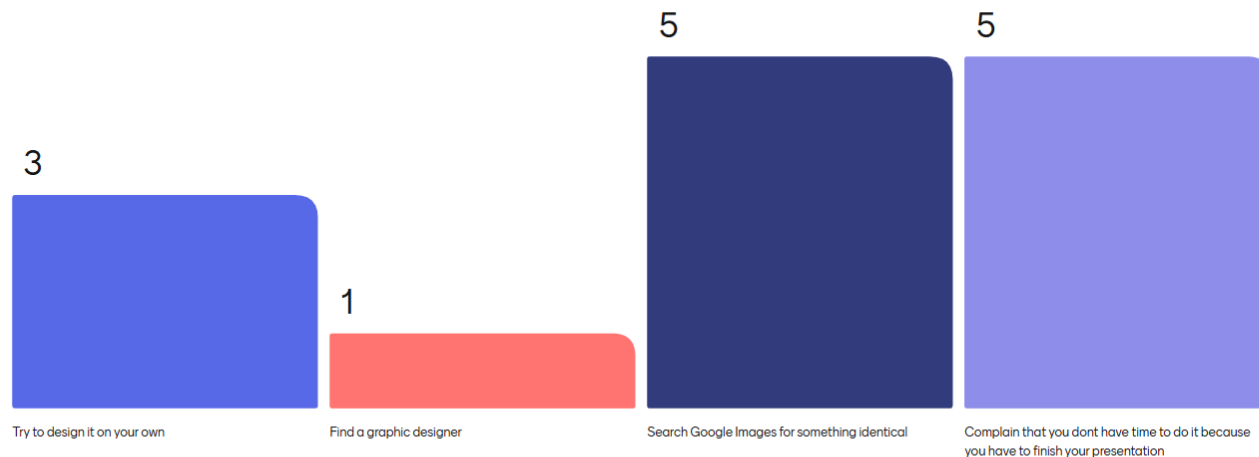
Join at menti.com | use code **2324 1884**



Menti
Lancaster



What is the first thing you do?



Choose a slide to present

- Good afternoon :)
- How much time (minutes) using each of the following tools?
Just your brain
Pen, paper & calculator
Computer & Excel
- How much time (minutes) using each of the following tools?
LLM & an Assistant
- Would you be able to build such AI without computers?



carsick dog sticker car



AI Mode

All

Images

Shopping

Videos

Forums

Short videos

More ▾

Tools ▾

Saved



Motion sickness



Slow driver



Bumper sticker



Vinyl sticker



Caution k9



On board



Jack russell



Driver carsick



Caution carsick



Etsy

Dog Car Sick - Etsy UK



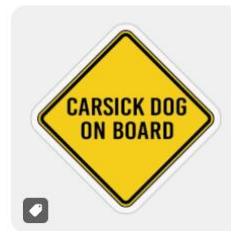
eBay

Vinyl Sticker Vehicle Bumper Decal | eBay



Amazon.com

StickerTalk Slow Driver Carsick Dog ...



Redbubble · In stock

Carsick Dog on Board fu...



Etsy

Dog Car Sick - Etsy UK



Etsy

Dog Car Sick - Etsy UK



eBay

10in x 3in Caution Carsick Dog on Board ...



Etsy

Dog Car Sick - Etsy



AliExpress

Waterproof Car Decal...



Amazon UK · In stock

LICHENGTAI Dog Car...



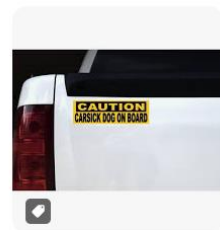
AliExpress

Waterproof Car Decal...



Joom

Yorkshire Dog Yorkie ...

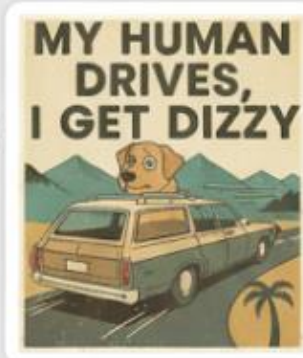


Walmart Business

Caution Carsick Dog o...

What if I can get personalized images?

Please create a car sticker design featuring the dog from the uploaded image. The dog should have a slightly sad or dizzy expression, similar to its current look. Place the dog inside a minimalist car interior, perhaps sitting in a back seat, with a visible window showing a blurred background to suggest motion. Include the text 'Carsick Dog - Please be patient!' prominently on the sticker. The overall style should be clean and suitable for a car decal.



Collaboration between Humans and GenAI

Statement (1): LLMs can provide significant productivity boosts for some tasks that the output can be easily verified.

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Statement (3): LLMs are good at generating unlimited content when prompted right. We are good at picking what feels good to us.

Question 4: How to collaborate effectively?

You would like to plan your summer holidays, but you don't have enough time because you have to prepare for 2 conferences.

You decide to ask help from your favorite LLM



Option 1

The Ever-Popular Beach Retreat: Sun, Sand, and Sea

For many, summer holidays are synonymous with relaxing on sun-drenched beaches. While the Greek islands remain a perennial favorite, a growing trend sees travelers exploring equally stunning and often more affordable coastlines just a stone's throw away.

Trending Destinations:

- **The Albanian Riviera:** Emerging as a chic and budget-friendly alternative to its more famous Mediterranean counterparts, the Albanian coast boasts turquoise waters, pristine beaches, and a vibrant nightlife. Popular spots include Saranda, Ksamil, and Dhërmi. The proximity to Greece makes it an easy and attractive option.
- **The Southern Italian Coast (Puglia & Calabria):** Accessible by ferry from Greece, the regions of Puglia and Calabria offer a captivating blend of dramatic coastlines, charming seaside towns, and delectable cuisine. Think the iconic trulli of Alberobello, the stunning beaches of Tropea, and the historic ports of Bari and Brindisi.
- **The Cyclades Islands, Greece (Paros, Naxos, and Milos):** While Mykonos and Santorini still draw crowds, the trend is shifting towards islands that offer a more authentic and relaxed experience. Paros is buzzing with a vibrant nightlife and beautiful beaches, Naxos is a haven for families and foodies with its long sandy shores and fertile interior, and Milos enchants with its unique lunar-like landscapes and secluded coves.

The Cultural Enthusiast's Dream: Cities Steeped in History and Art

For those who seek to immerse themselves in history, art, and vibrant urban life, a European city break offers a perfect blend of exploration and discovery. Direct flights from Athens open up a world of possibilities.

Trending Destinations:

- **Lisbon, Portugal:** With its colorful tiled facades, historic trams, and soulful Fado music, Lisbon continues to capture the hearts of travelers. Its affordability compared to other Western European capitals is a major draw.
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- **Budapest, Hungary:** The "Pearl of the Danube" beckons with its grand architecture, relaxing thermal baths, and a burgeoning culinary scene. A city of contrasts, it offers a dynamic and affordable European experience.

Option 2

1. The Peloponnese Peninsula, Greece

For an option closer to home that sidesteps the August island crowds, the Peloponnese Peninsula is an excellent choice. This vast and diverse region of the Greek mainland offers authentic experiences with far fewer tourists than the popular Cyclades.

- **Great Food:** The Peloponnese is a gastronomic treasure trove. It's renowned for its Kalamata olives and olive oil, citrus groves, and vineyards producing exceptional wines. You'll find rustic tavernas serving farm-to-table dishes, fresh seafood in the coastal villages, and unique regional specialties that you won't find elsewhere.
- **Sea:** The peninsula is fringed with stunning and often secluded beaches. From the sandy expanses of the western coast to the dramatic coves of the Mani Peninsula in the south, you can easily find a quiet spot to relax by the turquoise waters of the Ionian and Aegean Seas.
- **Avoiding Crowds:** While some historical sites like Olympia and Mycenae will have visitors, the coastline is extensive enough to allow for a sense of discovery and tranquility. Exploring the Mani or the coast of Messenia will lead you to charming stone villages and quiet beaches, even in the peak of summer.

2. Puglia, Italy

Located in the "heel" of Italy's boot, Puglia offers a more rustic and slow-paced Italian experience compared to the bustling Amalfi Coast. It's a region celebrated for its distinctive architecture, beautiful coastline, and outstanding culinary traditions.

- **Great Food:** Pugliese cuisine is a highlight of any trip to Italy. Known for its "cucina povera" (peasant cooking), the focus is on simple, high-quality local ingredients. Think fresh orecchiette pasta, creamy burrata cheese, an abundance of fresh vegetables, and fantastic seafood straight from the Adriatic and Ionian Seas. The region is also a major producer of olive oil and wine.
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Option 3

Option 1: Amorgos (The Authentic Cyclades)

- **Vibe:** Spiritual, raw, and traditional Greek. Unpretentious and quiet, with a focus on wild, natural beauty.
- **Landscape:** Dramatic, arid mountains and cliffs meeting the deep blue Aegean Sea. Classic whitewashed villages.
- **Best For:** Hiking with incredible sea views, exploring cultural sites (like the Hozoviotissa Monastery), and disconnecting in a rugged, soulful environment.
- **Food:** Hearty and traditional taverna fare (slow-cooked goat, fava, local cheeses). Excellent value.
- **Budget:** More affordable (€€).
- **Key Consideration:** Can be windy in August (the *Meltemi*), and the ferry from Athens is longer (6-9 hours).

Option 2: Paxos (The Lush Ionian)

- **Vibe:** Effortlessly chic, lush, and sophisticated yet relaxed. An intimate, "secret garden" feel with Italianate influences.
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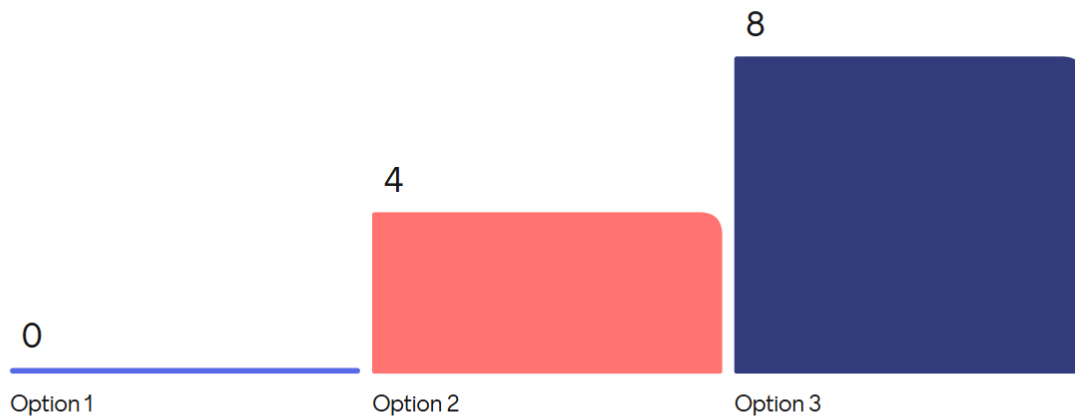
FT



Join at menti.com | use code **2324 1884**

 Mentimeter

Which of the LLM generated outputs would you find most actionable and helpful?



12 / 28

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Choose a slide to present

Good afternoon :)

How much time (minutes) using each of the following tools?

Just your brain

Plan, prepare & calculate

Computer & Excel

How much time (minutes) using each of the following tools?

LLM's and Python

Would you be able to build such AI without computers?

Option 1

Option 2

Option 3

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Option 1: Amorgos (The Authentic Cyclades)

A more advanced search engine.

With minimal context, expecting to magically do all the work

AI as a helpful assistant.

Providing a well-defined request with clear context and constraint

AI as a copilot/collaborator.

Breaking the problem down into a series of steps and iteratively guiding to a summarized solution

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oil and wine.

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Collaboration between Humans and GenAI

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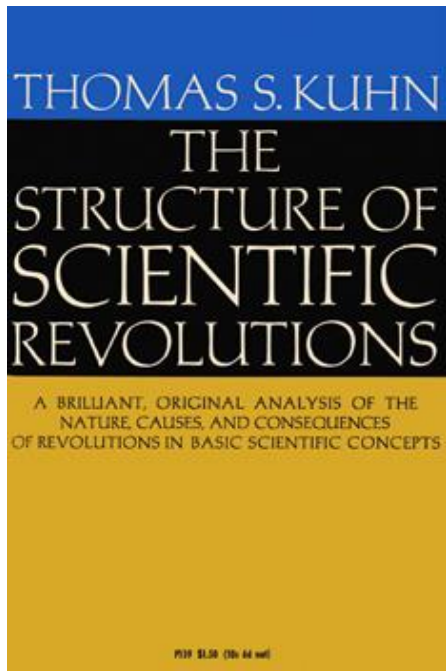
Statement (3): LLMs are good at generating unlimited content when prompted right. We are good at picking what feels good to us.

Statement (4): Collaboration is an iterative process. They generate, we adjust, they generate again. Finally, we get a summary of the collaborative solution.

Part 3:

Where we are currently at?

A two-wave model analogy.



Revolutionary Science:

(The Breakthrough Period)

Brief periods of paradigm-shifting breakthroughs.
(eg, ARPANET - first TCP/IP protocol- , Transistor)



We might have
passed this

Normal Science:

(The Consolidation Period)

Longer periods of problem solving with newly
established paradigms, working out implications,
refining methods.
(eg Internet, calculator computer etc)



We are probably
here

Where we are & where are we heading?

- Technology is pretty much established
 - Mostly efficiency breakthroughs

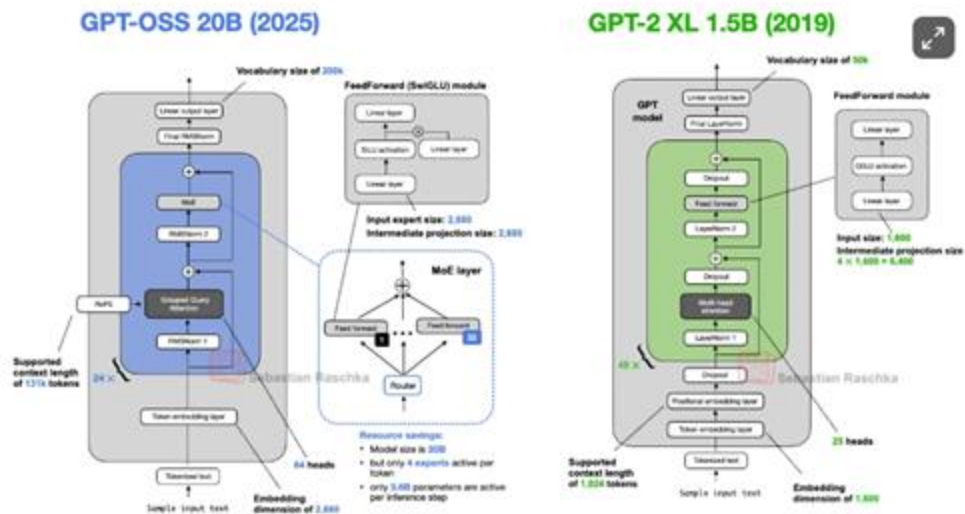


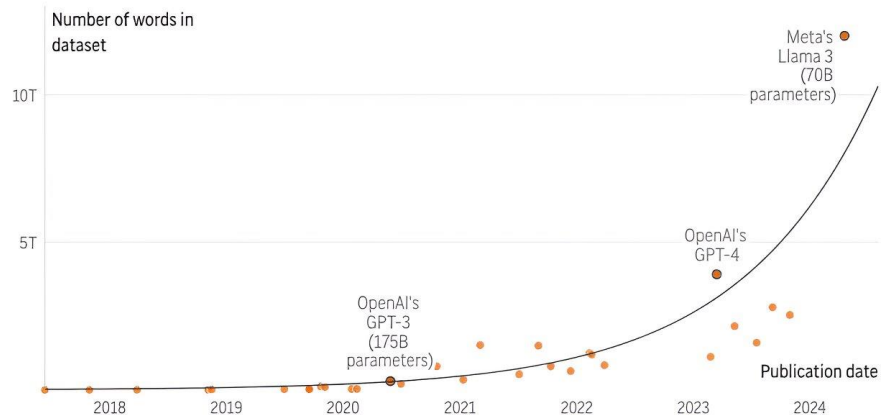
Figure 2: A side-by-side comparison between gpt-oss-20b and GPT-2 XL 1.5B.

Where we are & where are we heading?

- Technology is pretty much established
- We are running out of things to scale
 - Data 

LLM training datasets are growing

Datasets used to train key machine learning models have rapidly increased in size since 2017.



Source: Epoch AI

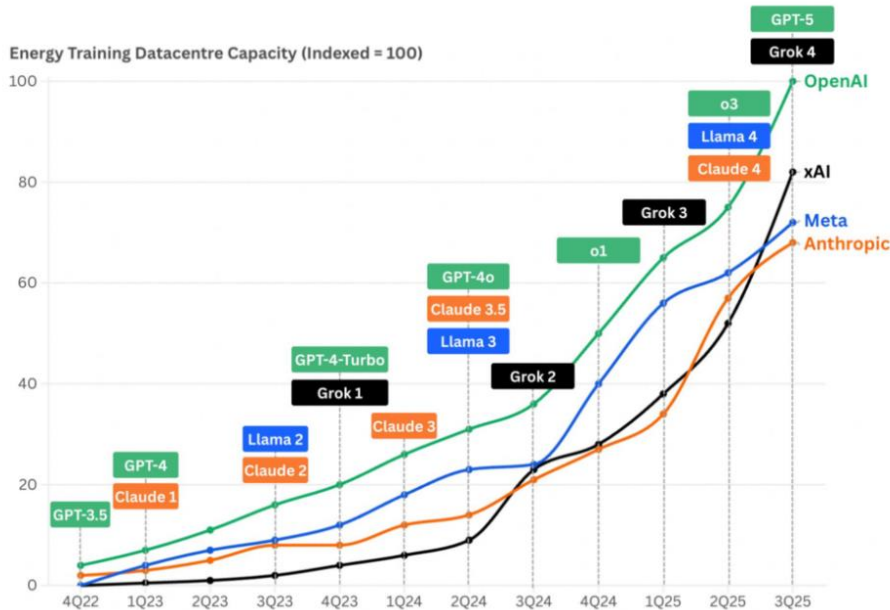
AP

Where we are & where are we heading?

- Technology is pretty much established
- We are running out of things to scale
 - Data ✓
 - Compute ✓

The amount of compute labs have has 3x in a year

OpenAI has the most compute* but xAI is growing the fastest



Where we are & where are we heading?

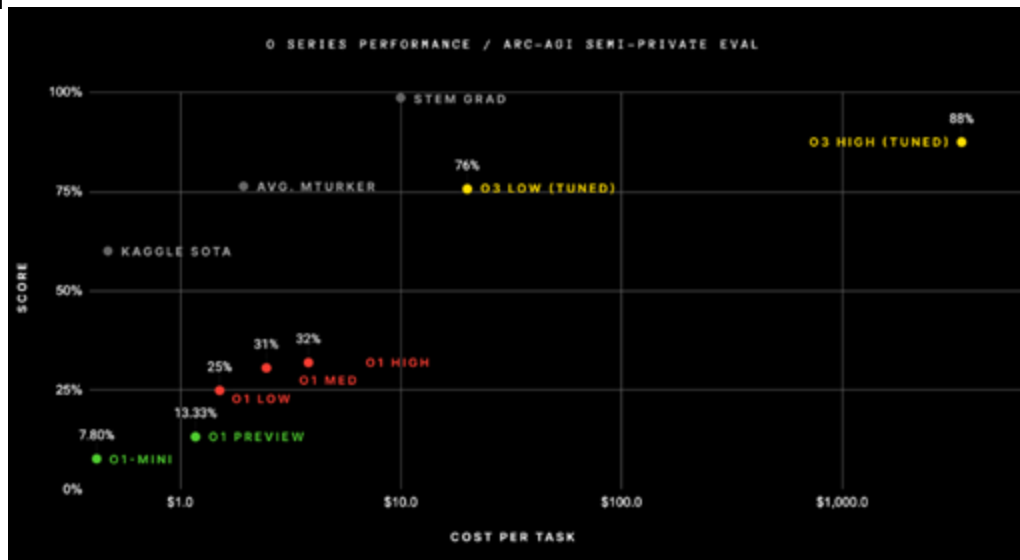
- Technology is pretty much established
- We are running out of things to scale
 - Data 
 - Compute 
 - Inference time (think longer)

DeepMind and OpenAI models solve maths problems at level of top students

For the first time, large language models performed on a par with gold medallists in the International Mathematical Olympiad.

Where we are & where are we heading?

- Technology is pretty much established
- We are running out of things to scale
 - Data ✓
 - Compute ✓
 - Inference time (think longer)



Where we are & where are we heading?

- Technology is pretty much established
- We are running out of things to scale
 - Data 
 - Compute 
 - Inference time (think longer)
 - New?

Where we are & where are we heading?

- Technology is pretty much established
 - We are looking for new things to scale
-



The Breakthrough Period

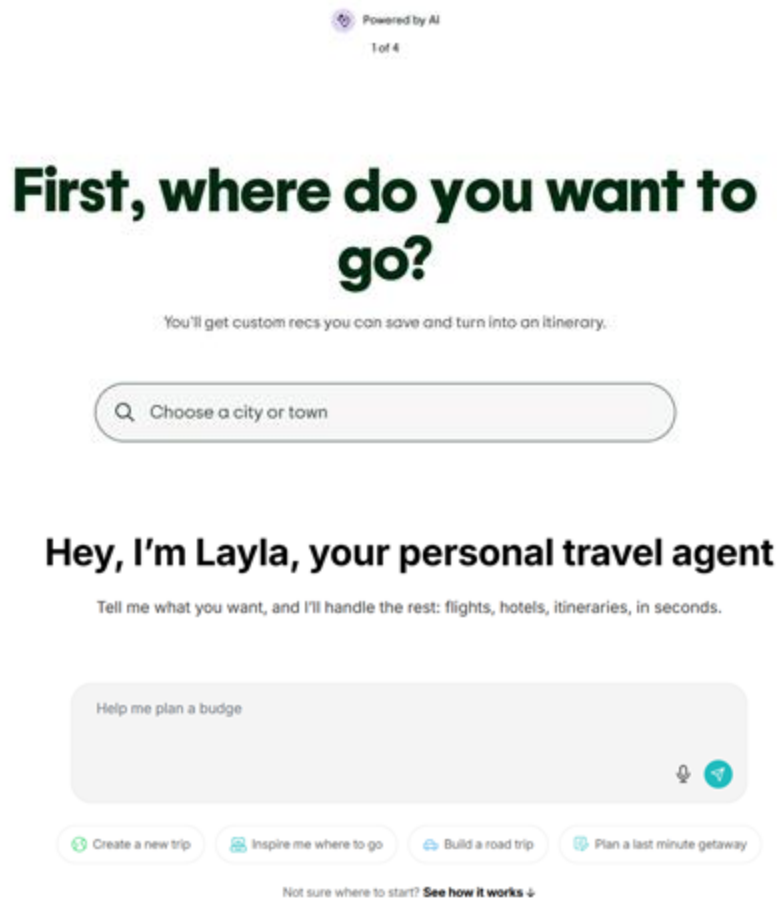
Where we are & where are we heading?

- Technology is pretty much established
 - We are looking for new things to scale
-
- We now have models for different tasks
 - Open/closed source
 - Different pricing
 - Different capabilities

Model	Input Price	Output Price	Context Length	Max Output Tokens	Arena Score
Google Gemini-2.5-Pro	\$2.50	\$15.00	1,000k	65k	1,456
OpenAI o3	\$10.00	\$40.00	200k	100k	1,444
OpenAI gpt-4o	\$2.50	\$10.00	128k	16k	1,443
OpenAI gpt-4.5	\$75.00	\$150.00	128k	16k	1,439
xAI Grok-3-Preview	\$3.00	\$15.00	131k	n/a	1,409

Where we are & where are we heading

- Technology is pretty much established
 - We are looking for new things to scale
 - We now have models for different tasks
-
- We now build AI Applications
 - Agentic applications.
 - For collaboration & task automation.
 - Using expertise injection



Where we are & where are we heading?

- Technology is established
 - Scaling dimension (more is better)
-
- Different model per task
 - New applications for task automation
 - Start figuring out the implications



The Breakthrough Period



The Consolidation Period

Part 4:

Implications in Research – Teaching - Admin

Adapting in the age of AI!

Please share your views!

Adapting in the age of AI?

Teacher group:

How would you adjust your teaching practises?

- Lecture material
- Workshops & labs
- Assignments
- Other?

Research group:

How can you improve your research process?

- Literature review
- Formulating research questions
- Experimenting & results analysis
- Other?

Engagement group:

How can you improve communication to non-academics?

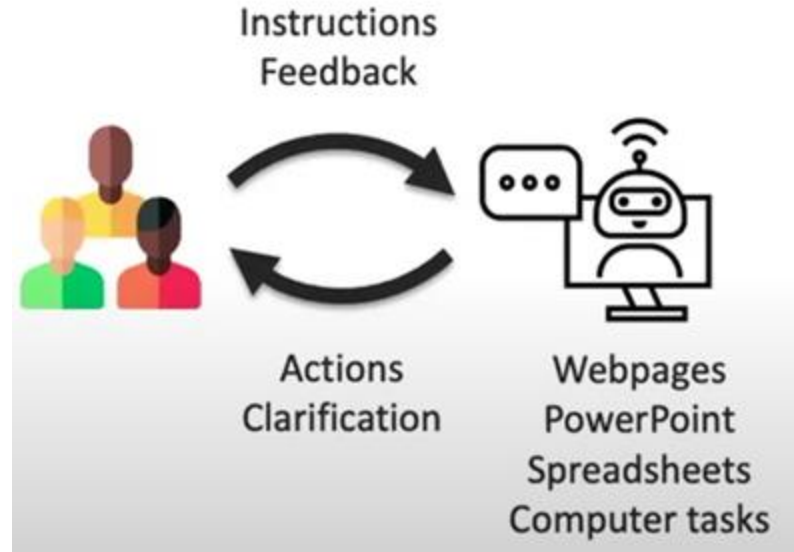
- Writing reports
- Compiling presentations
- Other?

Break 😊

Part 4:

Implications in Research – Teaching - Admin

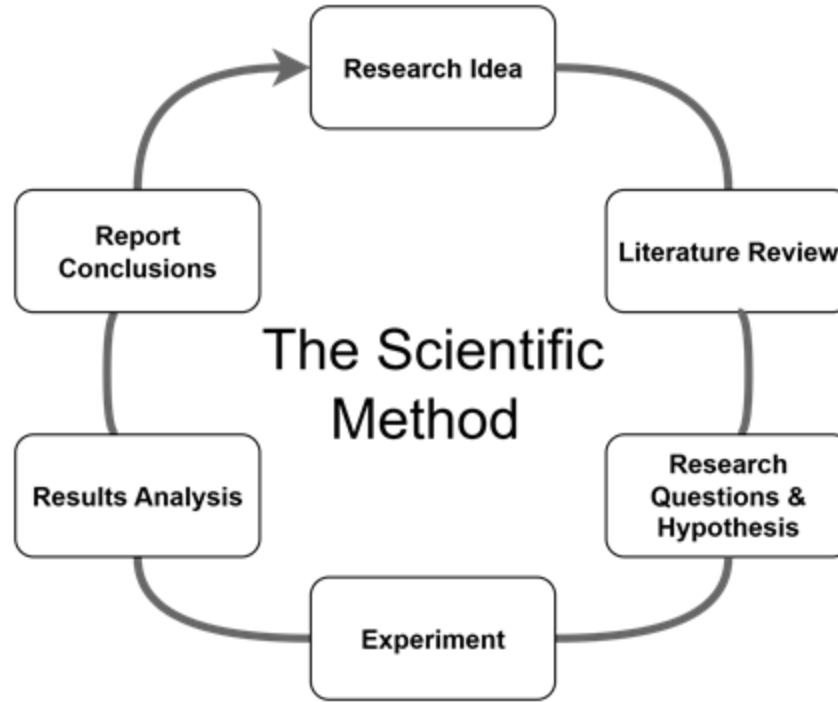
Remember: Augmentation not Replacement



Assumption & Direction:

GenAI is a (good) tool that we use collaborative to augment our (digital) capabilities

GenAI Implications in Research



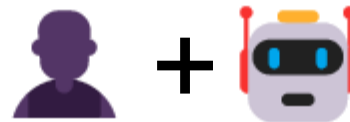
GenAI Implications in Research



- ✓ Maintain Full control
- ✗ Productivity enhancement
- ✗ Uneven competition



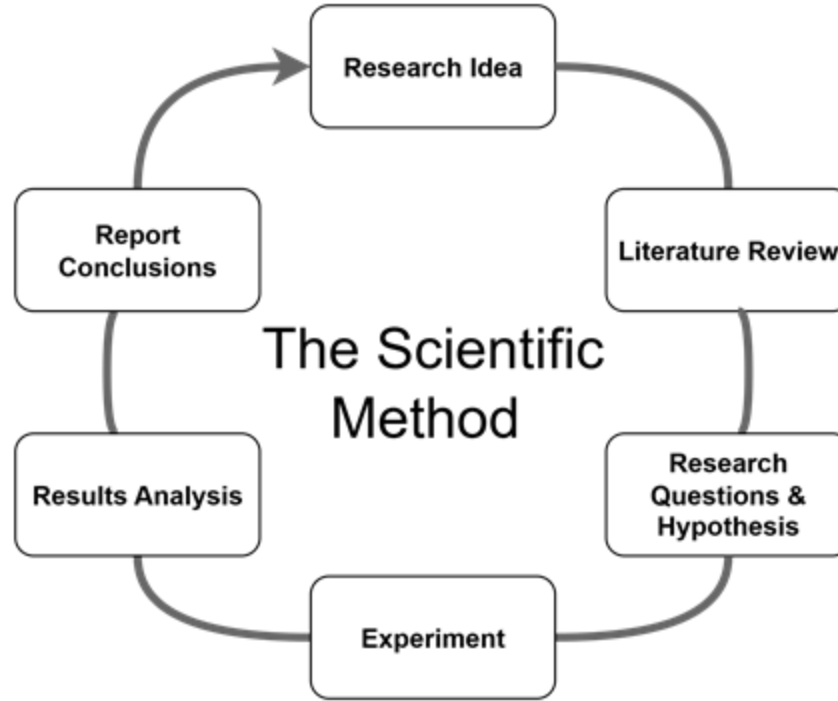
- ✓ Full automation
- ✗ Maintain control
- 😊 Have fun cleaning up



- 🤝 Collaboration
- 🔄 Automate trivial tasks
- ✓ Maintain control

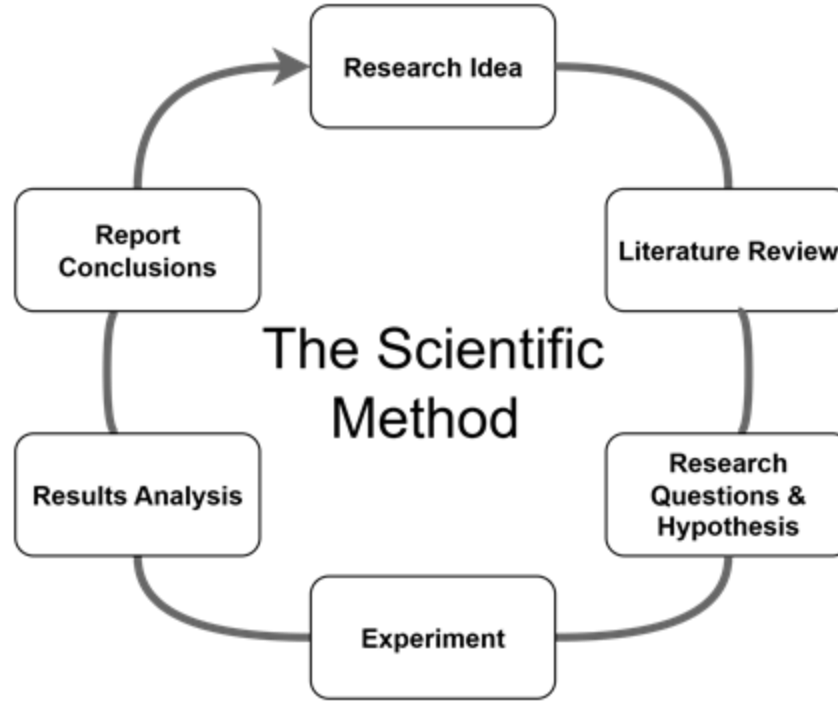
Experiment

GenAI Implications in Research



- ✓ Maintain Full control
- ✗ Productivity enhancement
- ✗ Uneven competition

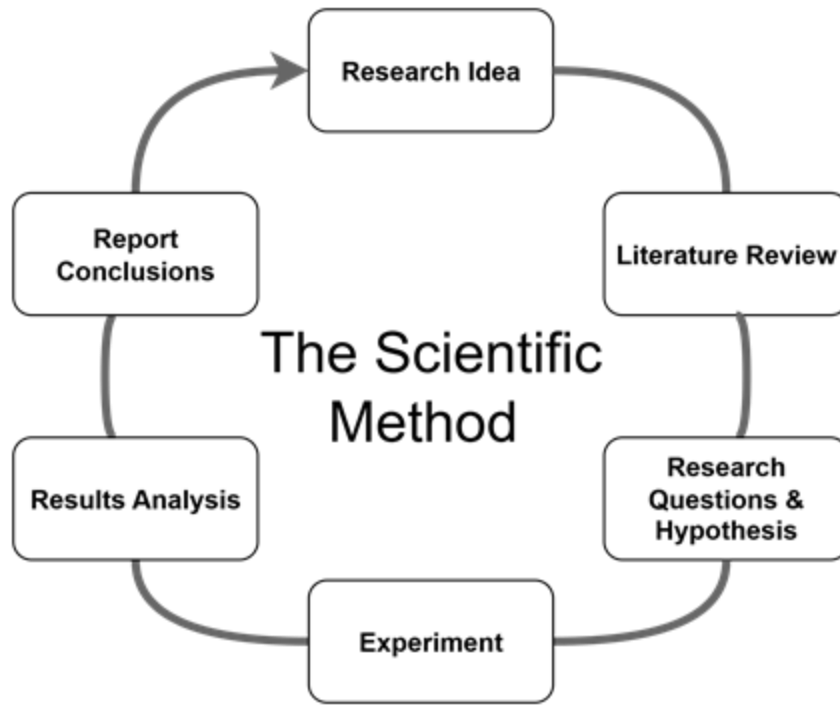
GenAI Implications in Research



- ✓ Full automation
- ✗ Maintain control
- 😊 Have fun cleaning up

Errors accumulate!

GenAI Implications in Research



Collaboration

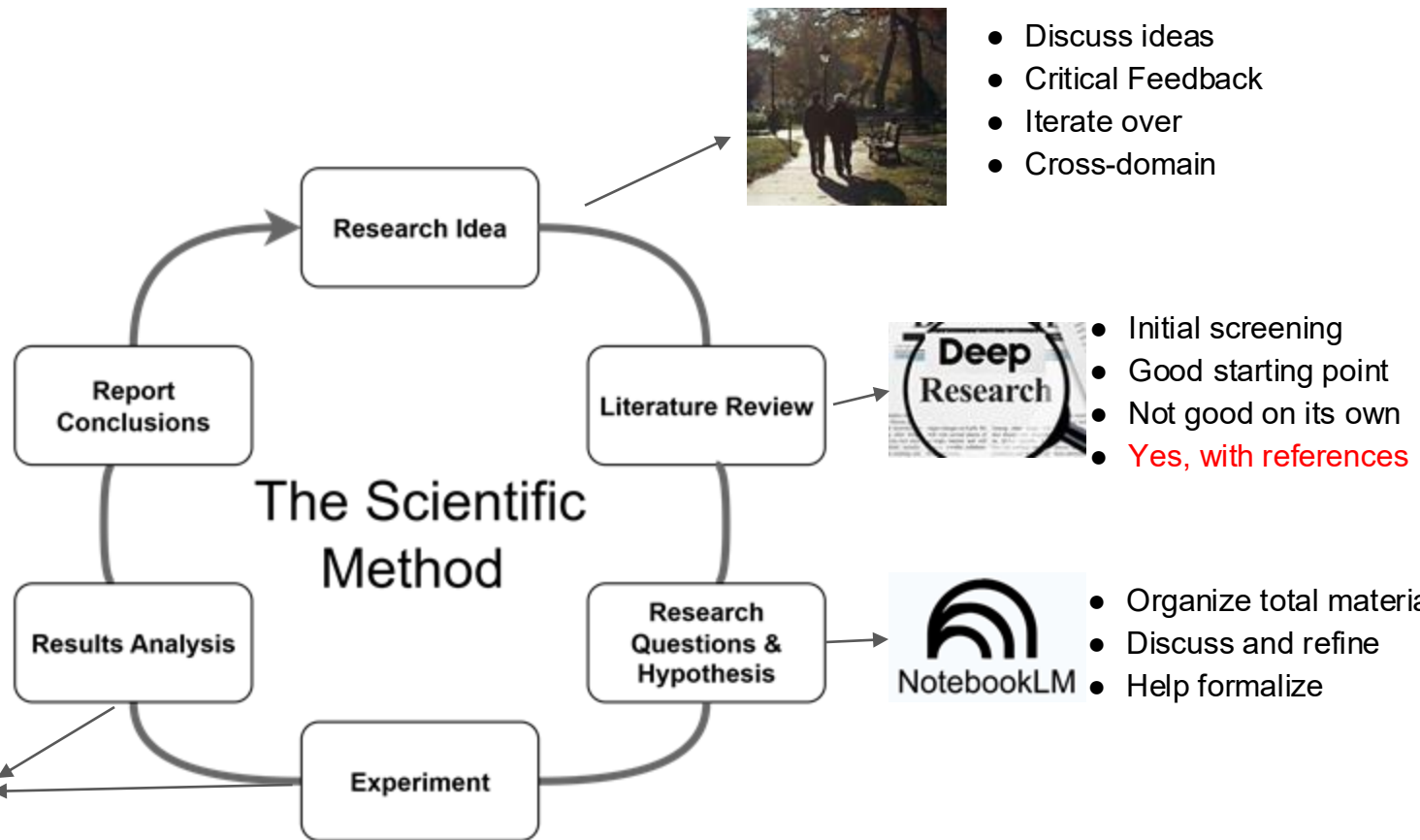


Automate trivial tasks

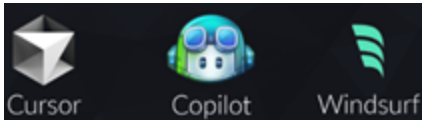


Maintain control

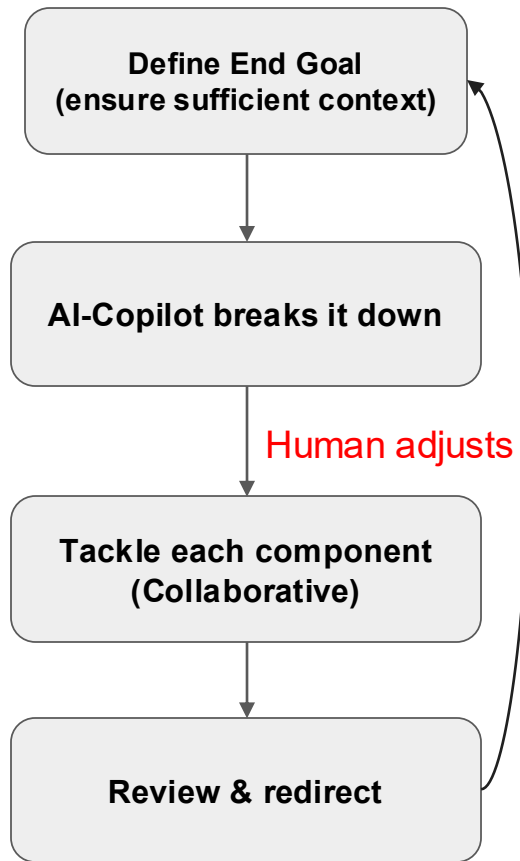
GenAI Implications in Research



Code specific models



GenAI Implications in Research



Andrej Karpathy

@karpathy

The hottest new programming language is English

Μετάφραση ανάρτησης

9:14 μ.μ. · 24 Ιαν 2023 · 8 εκ. Προβολές



1 χιλ.



7 χιλ.



48 χιλ.



4 χιλ.



```
threads.py 2 x instructions.py 9+
app > lib > threads.py > post_message
13 )
14 from .kv import kv
15 from .scenario_builder import generate_scenario
16
17 # Use the EventStreamSchema from models
18 event_stream_schema = EventStreamSchema
19
20 # Initialize OpenAI client
21 ai = AsyncOpenAI()
22
23 # Added logger instance
24 logger = logging.getLogger(__name__)
25
26
27 async def create_thread(
28     assistant_id: str,
29     difficulty: DifficultyLevel,
30     module_id: Optional[int] = None,
31     level: Optional[int] = None,
32 ) -> AsyncStream[AssistantStreamEvent]:
33     """Creates a new thread with an assistant, incorporating a generated scenario.
34
35     This function creates a new empty thread, generates a scenario based on
36     difficulty and optional module/level, stores the scenario in Redis, and then
37     initiates a run with the specified assistant using combined instructions
38     (base roleplay prompt + scenario details).
39
40     In the TypeScript version:
41     """
42     typescript
43     const create_thread = async (assistant_id: string, difficulty: DifficultyLevel) => {
44         const ai = new OpenAI();
45         const thread = await ai.beta.threads.create({ messages: [] });
46         const run = await ai.beta.threads.runs.create(thread.id, {
47             assistant_id: assistant_id,
48             additional_instructions: ROLEPLAY_INSTRUCTIONS(difficulty),
49             stream: true,
50         });
51     };
52
53 Roleplay agent instructions
54 Image instructions.py entry.txt
55 endpoints infrastructure prompts
56 read @entry.txt
57
58 I am testing the roleplay agent and the
59 instructions as given in @instructions.py
60
61 Look at that in the attached. If I reply something
62 stupid the agent goes to "How I can help you".
63 Why is the issue? Is it instructions. It does not
64 stick to the core instructions I provided
65
66 Restore checkpoint
67
68 Looking at your instructions.py file and
69 the issue you're describing, I can see the
70 problem. The roleplay instructions are not
71 being enforced strongly enough, and there's
72 a conflict between the general BASE_INSTRUC
73 TIONS and the specific roleplay behavior.
74 Let me analyze the current ROLEPLAY_INSTRUC
75 TIONS function and identify the issues:
76
77 Read instructions.py
78
79 Now let me look at the actual ROLEPLAY_INSTR
80 UCTIONS function:
81
82 Read instructions.py
83
84 4 files +130 -21
85 instructions.py endpoints
86 infrastructure prompts
87 Plan, search, build anything
88 Agent chat claude-4-sonnet
```

GenAI Implications in Teaching

We assume that students use GenAI for their studies

Digital Education Council Global AI Student Survey 2024 | Section 1. AI Usage



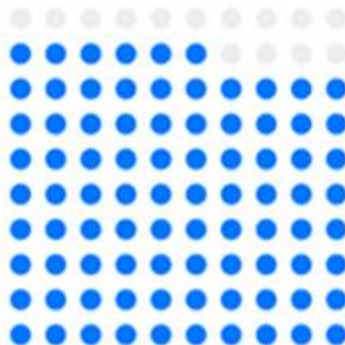
86% of students already use AI in their studies

Percentage of students using AI in their studies

Question: How often do you use AI tools?

86%

of students claim to use
AI in their studies

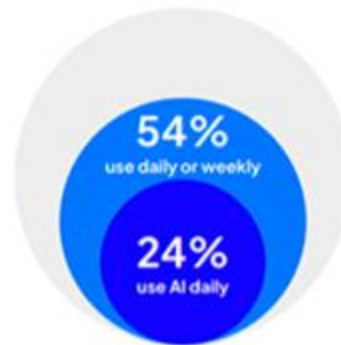


Frequency of students using AI in their studies

Question: How often do you use AI tools?

54%

of students use AI at least
on a weekly basis



Source: Digital Education Council Global AI Student Survey, 2024

GenAI Implications in Teaching: Challenges

1. Impact on learning process
2. Plagiarism
3. Evaluation
4. Curriculum development



AI usage for students

GenAI Implications in Teaching: Challenges

1. Impact on learning process
2. Plagiarism
3. Evaluation
4. Curriculum development



AI usage for students

Adaptation in the new reality seems the most promising approach



Addressing the challenges in 4 dimensions

Adaptation in the new reality seems the most promising approach

Clear guidelines

- No Permit vs Prohibit
- Scanning tools do not work
- Clear project guidelines
What, where, how.

Engaging Content

- Augment material
- Critical thinking
- Better hooks:
Attention is limited
- More personalized
AI tools for teachers!

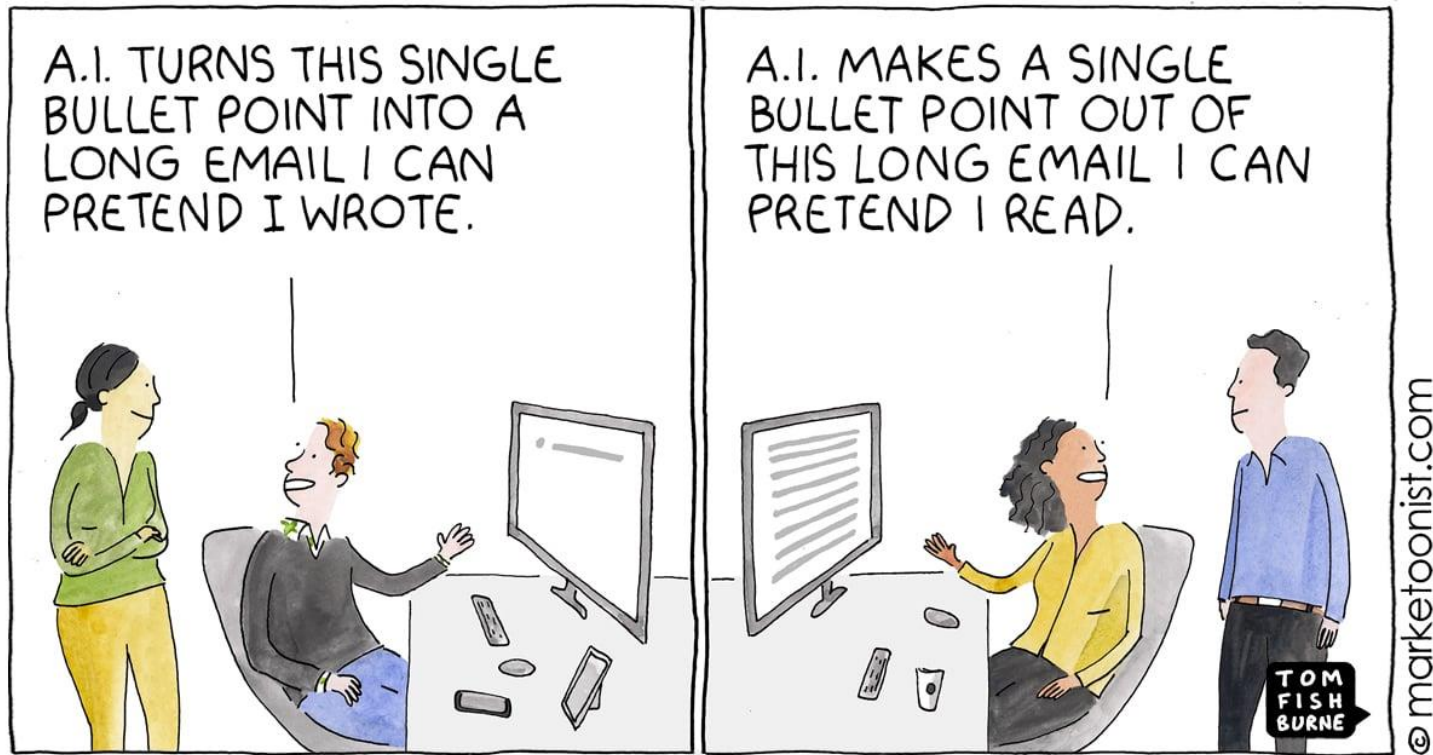
Assignments

- Reward processes
not products
- Break down tasks
into low stakes
- Ask for presentations

Educate usage

- Teach best practises
- Students should not
blindly trust output
- Seminars for good
practises

GenAI Implications in Engagement/Writing



GenAI Implications in Engagement/Writing

Note 1: End-to-end solutions rarely work

You
I have been invited by an association named "Early Career Forecasters" to give a talk on how I use AI technologies like chatGPT for research purposes.

Write the abstract and give a title to my talk.
The abstract should be 1 paragraph

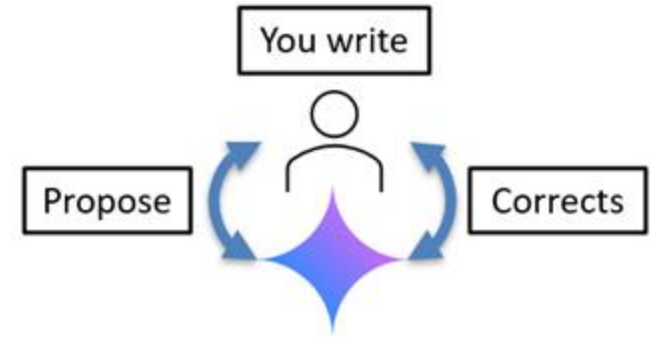
Plus, its not fun losing your writing style

Note 2: Setting the context is key.

```
You are [role/expertise].  
The audience is [specific group].  
The goal is [clear objective].  
Consider [constraints/requirements].
```

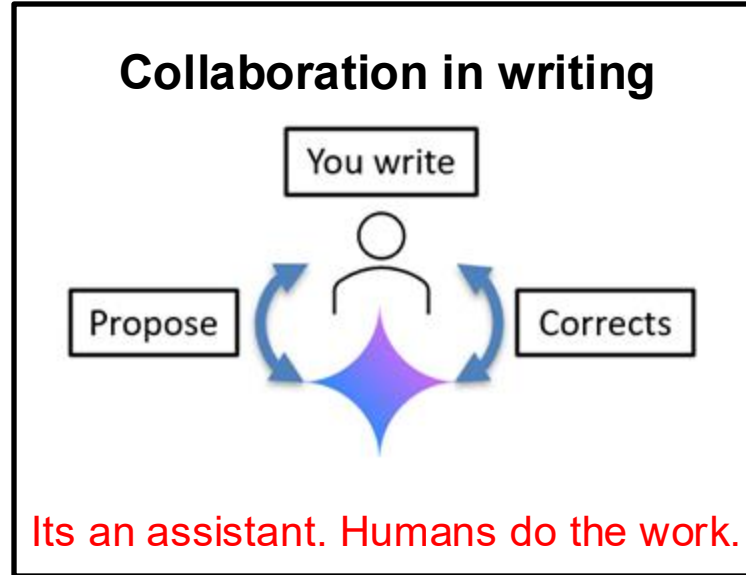
It narrows down the output space

Collaboration in writing

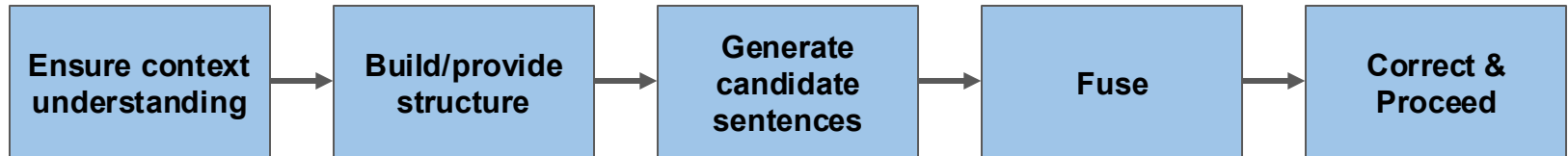


Its an assistant. Humans do the work.

GenAI Implications in Engagement/Writing



Note 3: Its an iterative circle with you in control



Putting everything together!

Effective collaboration is key



Verification & Guidance

- Context & Goals
- Quality Judgment
- Ethical Decisions
- Creative Direction



Content Generation

- Processing speed
- Infinite patience
- Pattern repetition
- Instruction following



Accountability - Transparency - Responsibility
(Ethical and responsible usage are a necessity for safe collaboration)

The Ironman Analogy

Augment based on the task at hand



Automated



Manual

Low stake tasks

A simple review is ok
(eg sentence rephrasing)

Medium stake tasks

Iterative human verification and action
(eg coding, literature review)

High stake tasks

Human executes the whole task
(eg grading students)

Some concluding remarks

- LLMs (and GenAI) are a powerful tool that assists us in problem solving.
- It's a technology that is here to stay. We are on the consolidation period.
- Understanding the implications and adapting should be a priority.
- Ethical and responsible collaboration is the most promising way forward.

Thank you for the attention 😊

Please share your questions, comments, concerns!



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<https://filtheo.github.io/>